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Steady Warps: Linear, Nonlinear, and Breaking

Jiaru Li [1] and Yoram Lithwick [1, 2]

[1] *Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA), Northwestern University, 1800 Sherman Ave, Evanston, IL 60201, USA*

[2] *Department of Physics and Astronomy, Northwestern University, 2145 Sheridan Rd, Evanston, IL 60208, USA*

ABSTRACT

An increasing number of protoplanetary disks shows observational signatures of warps and misalignments, raising questions of how disks sustain coherent warps and how they may break into misaligned pieces. We study the steady-state structures and breaking conditions of warped disks. To focus on the hydrodynamics, while remaining agnostic about what forces the warp, we adopt a simple but physically motivated setup: rather than including an explicit perturber or external torque, we fix the disk inclination angles $\beta_{\rm in}$ and $\beta_{\rm out}$ at the inner and outer boundaries. The disk is hence constrained to accommodate a warp between the boundaries. By varying the boundary misalignment $|\beta_{\rm out} - \beta_{\rm in}|$, we can explore the linear regime, the nonlinear regime, and the onset of breaking, while having good control over the warp amplitude. Combining this model with analytical theories and three-dimensional hydrodynamic simulations, we carry out a clean and systematic investigation of the hydrodynamic behaviors of warped disks. We find that, with small warps, disks settle into warp steady states that are well described by the linear theory. Moderately warped disks enter the nonlinear regime, showing several distinct features such as torque saturation, vertical "bouncing" motion of gas, and enhanced mass accretion rates. Measurements of these effects in our simulations show good quantitative agreement with nonlinear theories. Strongly warped disks are unstable: these disks are susceptible to a runaway growth of warp amplitude that ultimately leads to disk breaking. This instability may be caused by the nonlinear saturation of the disk internal torque, which occurs roughly when the warp amplitude exceeds a critical value $|\psi|_{\rm crit} \simeq 2\sqrt{\alpha}$ for Keplerian disks.

1. INTRODUCTION

Accretion disks are not always flat; gas circulating the same central object at different radii may have different orbital inclinations, producing warped structures.

Recent observations have found an increasing amount of evidence that such warps commonly exist in protoplanetary disks. In near-infrared scattered light images, many disks exhibit non-axisymmetric dark regions (e.g., Marino et al. 2015; Stolker et al. 2017; Benisty et al. 2018; Casassus et al. 2018; Muro-Arena et al. 2020; Keppler et al. 2020; Kraus et al. 2020), which is often interpreted as shadows cast by warps (see Benisty et al. 2023). Complementary evidence comes from molecular line observations, through which several disks are found to have large-scale $m = 1$ velocity structures consistent with warped or non-planar gas flows (e.g., Panić et al. 2010; Pineda et al. 2014; Casassus et al. 2015; Winter et al. 2025). In addition, increasingly many systems have been revealed to host both warps and multiple misaligned disk components, pointing to a rich dynamical history involving both disk bending and breaking (e.g., Kraus et al. 2020; Muro-Arena et al. 2020; Bohn et al. 2022). Together, these observations raise two fundamental questions: how do disks bend and sustain coherent warps, and under what circumstances do they break into misaligned components?

Theories of warped disks have been developed over the past several decades (see, e.g., Nixon & King 2016, for an overview). In these studies, a warp is formally defined as the radial derivative of the disk inclination. Misaligned disk annuli can exert internal torques on each other (due to pressure, etc), which in turn drives the changes of the inclination profiles of the whole disk.

Theories of warped disks have been developed over the past several decades (see, e.g., Nixon & King 2016, for an overview). In nearly Keplerian disks, neighboring disk annuli communicate through internal torques, causing warps to evolve through either bending-wave propagation or viscous diffusion, depending on the relation between α and the disk aspect ratio $h \equiv H/R$ (Papaloizou & Pringle 1983; Papaloizou & Lin 1995; Lubow & Ogilvie 2000). In the absence of forcing, free warps tend to damp away. In forced disks, however, the warp can relax to a steady configuration if the forcing evolves slowly enough (e.g., Lubow & Ogilvie 2000; Foucart &

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Lai 2014). Linear theory and simulations are in good agreement in this regime (e.g., Lodato & Pringle 2007; Lodato & Price 2010; Kimmig & Dullemond 2024; Fairbairn 2025).

For nearly-Keplerian disks, how warp evolves depends on the viscous parameter α (Shakura & Sunyaev 1973) relative to the disk aspect ratio $h \equiv H/R$. Linear theories show that, for $\alpha \lesssim h$, warps can propagate as bending waves at approximately half of the sound speed, while being viscously damped at a rate $\sim \alpha\Omega$, where Ω is the angular velocity of the gas (Papaloizou & Lin 1995; Lubow & Ogilvie 2000). When $\alpha \gtrsim h$, viscous damping becomes strong enough to suppress wave propagation, so the warp instead evolves in a diffusive manner (Papaloizou & Pringle 1983). As a result, in the absence of external torques, free warps tend to flatten on the intermediate timescale between the orbital period $\sim \Omega^{-1}$ and the global viscous timescale $\sim (\alpha h^2 \Omega)^{-1}$.

Warps in forced disks, on the other hand, may settle into steady states on the intermediate timescale if the relevant forcing acts more slowly (e.g., Lubow & Ogilvie 2000; Foucart & Lai 2014). Note that, if α is negligible, the evolution becomes sensitive to the difference between orbital and epicylic frequencies (i.e., the non-Keplerianity, see e.g., Ogilvie 1999, or Section 3). Hydrodynamic simulations have found good agreement with these linear theories (e.g., Lodato & Pringle 2007; Lodato & Price 2010; Kimmig & Dullemond 2024; Fairbairn 2025).

The nonlinear regime, where warp amplitudes are large, is more complex and less well understood. Analytical theories predict several important nonlinear effects, such as modifications of the internal torques (Ogilvie 1999; Ogilvie & Latter 2013a; Dullemond et al. 2022), changes to the mass accretion flow (e.g., Pringle 1992; Ogilvie 1999), and the so-called "bouncing effect", in which gas elements would perform rapid vertical compression and expansion as they orbit around the star (Ogilvie & Latter 2013a; Fairbairn & Ogilvie 2021a,b; Held & Ogilvie 2024). These features have been identified in several numerical simulations (e.g., Lodato & Price 2010; Sorathia et al. 2013; Deng & Ogilvie 2022; Kaaz et al. 2023, 2025; Kimmig & Dullemond 2024). In low-α disks, wave coupling can also lead to parametric instability (Gammie et al. 2000; Ogilvie & Latter 2013b), which was not seen in the early numerical studies but emerged in recent high-resolution simulations (e.g., Paardekooper & Ogilvie 2019; Deng et al. 2021; Deng & Ogilvie 2022; Fairbairn & Stone 2025). Due to the complex nature of nonlinear hydrodynamics, recent studies tend to either pursue more powerful simulations (e.g., Deng & Ogilvie 2022; Kaaz et al. 2025) or develop alternative frameworks, such as affine models (Ogilvie 2018) and ring models (Fairbairn & Ogilvie 2021a).

Extreme nonlinear evolution can lead to disk breaking, where the disk splits into multiple disconnected planes (e.g., Larwood et al. 1996; Nixon et al. 2012, 2013; Zhu 2019; Liska et al. 2021). Several theories have been proposed to explain the mechanisms of breaking, including resonance with tidal forcing (Lubow & Ogilvie 2000; Martin et al. 2020), rapid nodal precession (Zhu 2019; Martin et al. 2020; Rabago et al. 2024), and linear instabilities related to the anti-diffusion of the disk density (Doğan et al. 2018; Raj et al. 2021). Yet a general condition and physical mechanisms that trigger disk breaking remain unclear.

In this paper, we focus on the long-lived warp steady state (WSS), the configuration in which rapid bending-wave oscillations have damped out and the disk's inclination profile stops changing, despite the slow evolution set by the changing background surface density. Such states are more likely to be observable than their transient predecessors: given the lifetimes of protoplanetary disks, any warps present are likely to have relaxed into quasi-steady configurations. Our primary goals are to determine the WSS structures of disks, both in the linear and the nonlinear regimes, and to diagnose how a sufficiently large warp drives disk breaking.

We perform both theoretical analysis and numerical simulations. To focus on the hydrodynamics, while remaining agnostic about what forces the warp, we adopt a simple but physically motivated setup: rather than include an explicit perturber or external torque, we fix the disk inclination angles $\beta_{\rm in}$ and $\beta_{\rm out}$ at the inner and outer boundaries. The disk is hence constrained to accommodate a warp between the boundaries, and we can study the resulting steady-state structure. In the simulations, we also lower the surface density in the middle of the disk (but keep it non-empty), allowing us to localize the warp and reduce the possible influence of artificial boundary conditions. Our setup provides a clean laboratory for testing the hydrodynamic response of disks to generic perturbations. By varying the boundary misalignment $|\beta_{\rm out} - \beta_{\rm in}|$, we can explore the linear regime, nonlinear regime, and the onset of breaking, while having good control over the warp amplitude. As we will show, our simulations show excellent agreement with the theories.

The rest of this paper is organized as follows. We review the linear theory of warped disks in Section 3, and test the theory using hydrodynamic simulations in Sections 3.3. In Section 5, we extend the theory in nonlinear regime, allowing us to estimate the maximum mu-

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tual inclination $|\beta_{\rm out} - \beta_{\rm in}|$ for which a disk can sustain a continuous warp. We then perform hydrodynamics simulation in Section 6 to analyze the nonlinear behaviors and the breaking process of disks when $|\beta_{\rm out} - \beta_{\rm in}|$ becomes large. Finally, we conclude in Section 8.

2. INCLINATION AND WARP PROFILES

We quantify the inclination profile of a disk via the direction of the angular momentum vector per unit radius,

$$\boldsymbol{L}(r) \equiv \int \rho \boldsymbol{r} \times \boldsymbol{v}\, r^2 d\Omega \tag{1}$$

where r is radial distance to the star, $d\Omega$ is the element of solid angle, and ρ and $\boldsymbol{v}$ are the density and velocity fields. The unit vector

$$\hat{\boldsymbol{l}}(r) \equiv \boldsymbol{L}/|\boldsymbol{L}| \tag{2}$$

is used to define the inclination relative to a reference plane, which we take to be at $z = 0$. In particular, we parameterize the Cartesian components as

$$\hat{\boldsymbol{l}} = [\sin\beta\cos\gamma, \sin\beta\sin\gamma, \cos\beta] \ , \tag{3}$$

which implicitly defines the inclination (β) and the longitude of ascending node (γ, after a shift by $\pi/2$). We shall use $\hat{\boldsymbol{l}}$ interchangeably with $\{\beta, \gamma\}$.

The "warp" $\boldsymbol{\psi}$ is defined as the radial derivative,

$$\boldsymbol{\psi} \equiv r\partial_r \hat{\boldsymbol{l}} \ . \tag{4}$$

A flat disk has $\boldsymbol{\psi} = 0$, whether or not it is inclined. A warp can have two components: a "bend" (from $\partial_r\beta$), and a "twist" (from $\partial_r\gamma$). See Figure 1 for a cartoon of a warped disk with no twist. For the disks studied in this paper, the twist is much smaller than the bend.

3. LINEAR INCLINATION PROFILE

3.1. *Linear Equations of Motion*

We derive the linearized equations that govern a warp in Appendix A. We take the unperturbed disk to coincide with the $z = 0$ plane, and perturb the Navier-Stokes equations to linear order in inclination (β). We assume a globally isothermal equation of state, time-independence (i.e., wave steady state or "WSS"; see Introduction), and adopt a Shakura-Sunyaev viscosity parameter α. For the perturbed variables, we set their azimuthal dependence $\propto e^{-i\phi}$ (i.e., Fourier with $m = 1$, corresponding to an inclined disk), and their vertical dependence to be the leading term in a Hermite expansion. The resulting equations are very simple, and are expressed in terms of the complex-valued Fourier-Hermite amplitudes of the velocity field (Equations A48–A49).

Although Equations (A48)–(A49) are sufficient for considering the linear evolution, in this paper we shall also model nonlinear evolution. With that goal in mind, it is more convenient to express the equations in terms of vectors rather than complex amplitudes. We do that conversion in Appendix B. The resulting equations are

$$4\alpha\boldsymbol{U}_\perp + 2\epsilon\hat{\boldsymbol{l}} \times \boldsymbol{U}_\perp = \boldsymbol{\psi} \quad \text{("sloshing equation")} \tag{5}$$

$$\boldsymbol{G}_\perp = -\sigma\boldsymbol{U}_\perp \tag{6}$$

$$\partial_r \boldsymbol{G}_\perp = 0 \tag{7}$$

where ϵ quantifies the non-Keplerianity of the disk (Equation A16), and σ is a scaled surface density,

$$\sigma(r) \equiv 2\pi\Sigma H^2 r^2 \Omega^2, \tag{8}$$

which is expressed in terms of surface density (Σ), scale height (H), and angular velocity (Ω). Given the σ profile and boundary conditions, these equations may be solved to obtain the $\boldsymbol{\psi}$ profile, and thence the $\hat{\boldsymbol{l}}$ profile from Equation (4). We discuss the other dependent variables, $\boldsymbol{U}_\perp(r)$ and $\boldsymbol{G}_\perp(r)$, shortly.

In Section 4 we provide a detailed discussion of the derivation and meaning of these equations. Here we provide a brief overview. Equation (5) quantifies how the warp drives sloshing, which refers to velocities perpendicular to $\hat{\boldsymbol{l}}$. The amplitude and phase of the sloshing is encoded in the vector $\boldsymbol{U}_\perp$, where we use a subscript $\perp$ to indicate that the vector is perpendicular to $\hat{\boldsymbol{l}}$. Equation (6) states that sloshing drives a torque, $\boldsymbol{G}_\perp$, which is defined in terms of primitive quantities in Equation (14). And Equation (7) states that the torque must be constant, because torque is the flux of angular momentum, and angular momentum cannot build up in steady state.

Equations (5)–(7) are equivalent to Equations (12) and (13) in Lubow & Ogilvie (2000), after setting $\partial_t = 0$ and dropping their external torque. These linearized warp equations were first derived by Papaloizou & Pringle (1983) and Papaloizou & Lin (1995) including time dependency. In general, different equations apply when $\alpha < H/r$ and $\alpha > H/r$, which are called, respectively, the "wave-like" and "diffusive" regimes (see Nixon & King 2016, for a review). But for the time-independent equations, it is well known that Equations (5)–(7) hold in both regimes, as is also evident from our derivation in Appendix A. Hence we need not distinguish those two regimes in this paper.

3.2. *Linear Solutions*

Equations (5)–(7) are solved in, e.g., Lubow & Ogilvie (2000) and Foucart & Lai (2014). We solve them here by setting $\hat{\boldsymbol{l}} \to \hat{\boldsymbol{z}}$ and taking the $\perp$ vectors to lie in the x-y

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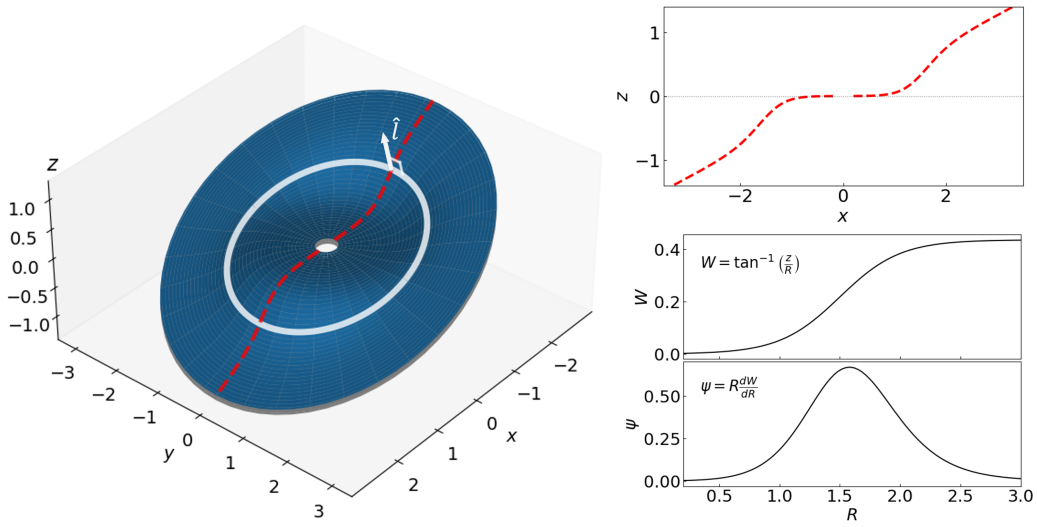


Figure 1. **Left:** Cartoon of a warped disk. For this example, there is a warp, but no twist ($\gamma = 0$). The red dashed curve traces the orientation: it is the line of nodes, after shifting by $\pi/2$. The white circle is a fiducial radius, and the arrow $\hat{\boldsymbol{l}}$ is the normal to the disk plane at that radius. **Upper right:** Same red-dashed curve as in the left panel, showing its x-z coordinates. **Lower right:** Profiles of the inclination angle β and the warp.

plane, valid to linear order. We then set $\boldsymbol{G} \to$constant (from Equation 7), and insert $\boldsymbol{U}_\perp \to -\boldsymbol{G}_\perp/\sigma$ in Equation (5), resulting in $-(4\alpha + 2\epsilon\hat{\boldsymbol{z}}\times)\boldsymbol{G}_\perp/\sigma = \boldsymbol{\psi}$. We convert from vectorial to complex form[1] by taking the dot product of that equation with $\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}$, yielding

$$\frac{4\alpha + 2i\epsilon}{\sigma} \times \text{const} = r\partial_r\left(\beta e^{i\gamma}\right), \tag{9}$$

where the constant on the left-hand-side is the complexified $-\boldsymbol{G}_\perp$, i.e., const$=-\boldsymbol{G}_\perp \cdot (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})$, and on the right-hand side $\beta e^{i\gamma} \approx \hat{\boldsymbol{l}} \cdot (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})$, which is the usual complex inclination.

For a *free* disk, in which the boundaries are not torqued externally, the constant in Equation (9) must vanish ($\boldsymbol{G}_\perp = 0$). The solution is then $\beta e^{i\gamma} =$const., i.e., the steady disk must be a flat plate.

In our setup, the inner and outer inclinations are externally imposed, and so the constant in Equation (9) is determined by the difference between inner and outer inclinations. The derivative on the right-hand side of that equation is the (complexified) warp, i.e., $r\partial_r(\beta e^{i\gamma}) \approx \boldsymbol{\psi} \cdot (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})$, to linear order, and so the equation shows that the warp is largest where the surface density (and hence σ) is smallest, ignoring ϵ for now. Intuitively, a smaller surface density means that the disk is weaker, and the disk bends primarily where it is weakest. The inclination profile is obtained by integrating Equation (9):

$$\beta e^{i\gamma} = \text{const} \times \int_{r_{\rm in}}^{r} \frac{4\alpha + 2i\epsilon(r')}{\sigma(r')} \frac{dr'}{r'}, \tag{10}$$

where we assume without loss of generality that the inner inclination is $\beta = 0$. The constant in Equation (10) is determined by setting the inclination at the outer edge to its desired value.

Figure 2 shows the inclination profiles that result from using three different Σ profiles in Equation (10). We set the domain to be

$$0.5 < r < 3$$

and

$$\alpha = 0.019,$$
$$H = 0.02 \times r^{3/2},$$

where the latter is needed to obtain ϵ (Equation A16). We set the outer inclination to $\beta_{\rm out} = 1$. In the linear limit, the inclination profile scales with $\beta_{\rm out}$.

The three Σ profiles are shown in panel (a). The green one is a power law, $\Sigma \propto r^{-3/2}$. The resulting β and γ are shown in panels (c) and (d). The warp (panel (e)) is largest at the inner boundary. In general, a power-law Σ profile produces a warp that is largest at a boundary. For the other two Σ profiles, we impose gaps of depth 0.2 and 0.05. As a result, the warp is concentrated in the gap. For our simulations below, it will prove convenient to keep the warp away from the domain boundaries, to

[1] Converting to complex form makes the mathematical solution more transparent. In truth, we derive Equations (5)–(7) in complex form in Appendix A, and then convert them to vectorial form in Appendix B. Hence converting them back to complex form is somewhat circular. We adopt this approach because we prefer to organize the discussion in the body of the paper around the vectorial equations, as they are more readily extendible to the nonlinear case.

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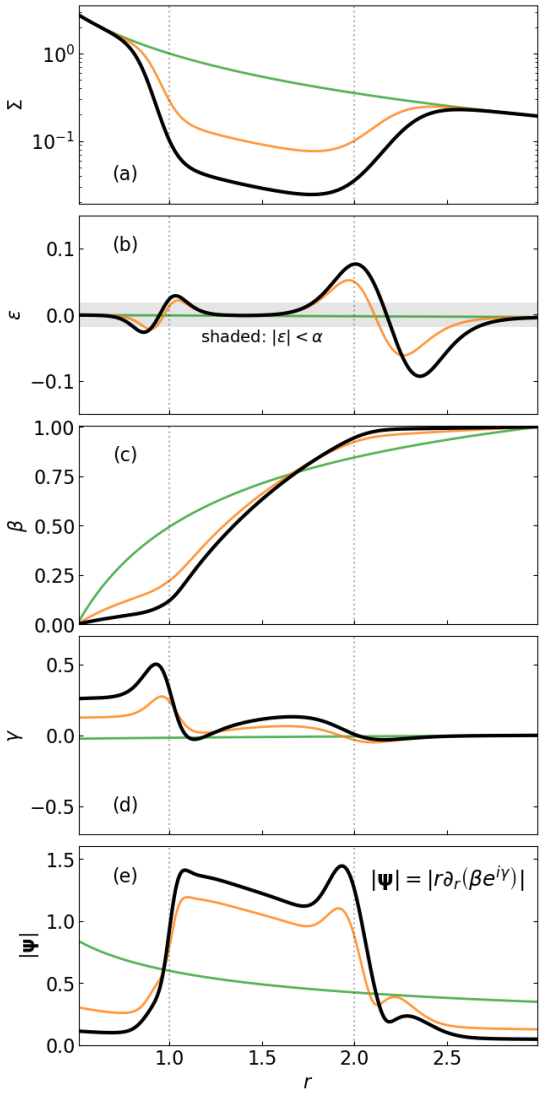


Figure 2. Linear warp in steady state, for disks with different density profiles. **Panel (a)**: Input surface density profiles; vertical dotted lines indicate roughly the edges of the density gap. **Panel (b)**: Non-Keplerianity parameter ϵ calculated based on Σ. The shaded area shows where $|\epsilon| < \alpha$. **Panels (c)-(e)**: Inclination and warp profiles, calculated from Equation (10).

avoid unphysical boundary effects. We will do that by imposing a gap. Imposing a gap introduces a minor complication: ϵ can get large at gap edges, as shown in the lower-left panel. Nonetheless, as long as $|\epsilon| \ll \alpha$, as is true within most of the gap for the profiles shown, the effect of ϵ may be ignored there.

3.3. *Simulation of a Linear Warp*

We simulate the 3D hydrodynamics of warped disks with Athena++ (Stone et al. 2020), which solves the Navier-Stokes equations for a globally isothermal, viscous disk in spherical polar coordinates (r, θ, ϕ). Implementation details are discussed in Appendix D.1. Our simulation setup mimics that of Section 3.2: the initial surface density profile is the same as the black curve in Figure 2, with a gap in the central region. The disk is then evolved while holding the inclinations at the inner and outer boundaries fixed. The inner inclination is held at zero, which implies that the polar/z-axis of the simulation is perpendicular to the inner disk (as in Figure 1). The outer inclination is held at $\beta_{\rm out}$. In the present section, we set $\beta_{\rm out} = 0.033 \approx 2°$, well within the linear regime. We analyze higher values of $\beta_{\rm out}$ in later sections. Table 1 lists the main simulation parameters. The time unit is chosen so that the Keplerian angular frequency at $r = 1$ is unity.

We use simulation C03 to verify the linear prediction for the inclination profile (Equation 10). Initially, the inclination profile in C03 is chosen to rise linearly across the domain. Snapshots of Σ, β, and γ at $t = 0$ and two subsequent times are shown in Figure 3. We measure $\beta(r)$ and $\gamma(r)$ from the simulations via Equations (1)–(3).[2]

The surface density evolution (panel a) shows that the initial gap fills in viscously. The evolution of Σ is virtually unaffected by the warp in this run, as we have confirmed by comparing with a zero-inclination disk. The viscous time across a distance r is $t_{\rm vis} \approx \alpha^{-1} h^{-2} r^{3/2} \approx 10^5 r^{1/2}$. The evolution timescale of Σ seen in Figure 3(a)

Table 1. Simulation parameters. The upper table lists parameter values that are common to all runs. The lower table summarizes run-specific parameters: the outer inclination $\beta_{\rm out}$, the half-width of the domain in θ, and the number of grid points. As described in the text, C03 has a different initial inclination profile than the others.

c_s	α	r-domain
0.02	0.019	(0.5, 3.0)

Name	$\beta_{\rm out}$	half-width in θ	(N_r, N_θ, N_ϕ)
C03	0.033	0.24	(256, 256, 260)
C10	0.10	0.74	(256, 728, 260)
C15	0.15	"	"
C20	0.20	"	"
C30	0.30	"	"
C40	0.40	"	"

[2] We measure the (angle-averaged) surface density profile via

$$\Sigma(r) = \frac{1}{2\pi r} \int \rho r^2 d\Omega \ .$$

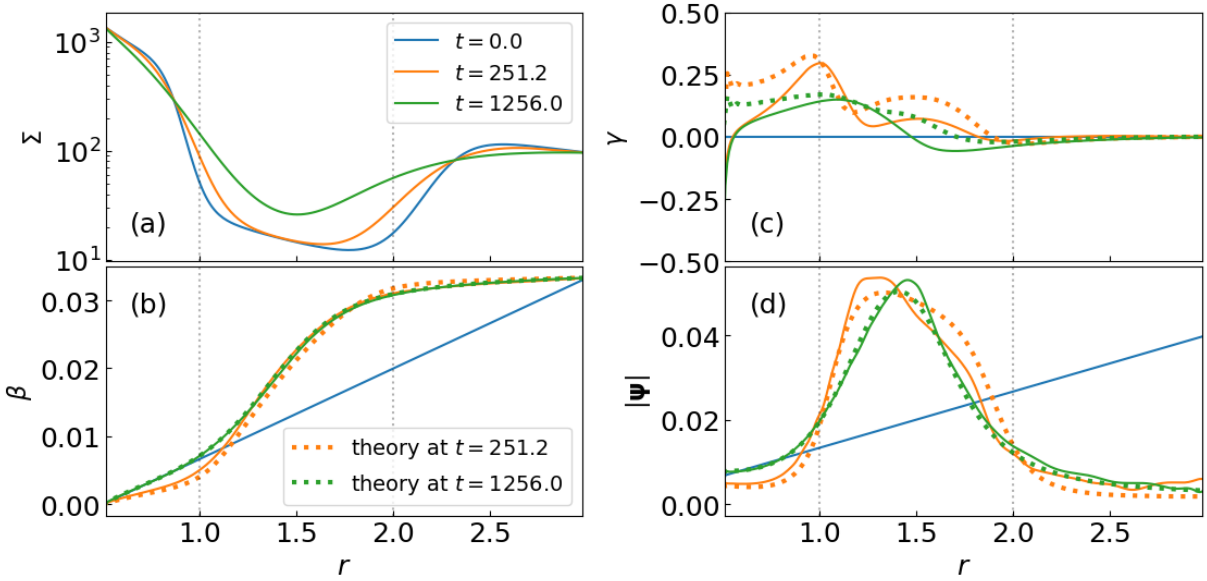


Figure 3. Snapshots of profiles in a weakly warped disk (simulation C03). **(a)** Surface density; **(b-c)** Inclination angles; **(d)** Warp. Solid curves represent simulation data, and dotted curves show the linear prediction (Equation 10), using the simulated Σ and ϵ.

is faster than that by a factor of around 100, due to the narrowness of the gap edges.

The time for the inclination to reach its steady profile is much shorter than the Σ evolution timescale. There are two relevant warp timescales, one associated with the time for a warp to travel $t_{\text{warp},c_s} \approx (2r/c_s) = 100r$, and the other for the warp to damp $t_{\text{warp},\alpha} \approx \alpha^{-1} r^{3/2} = 52r^{3/2}$. Figure 4 shows the time evolution of β at the middle of the gap. The initial evolution exhibits a damped oscillation, on a timescale of ~ 100, in agreement with the two warp timescales. After that, the inclination evolves much more slowly, on the viscous timescale of Σ, suggesting that the disk is in a warp steady state. Therefore beyond a time of ~ 100, the steady warp theory (Section 3) should be applicable.

Figure 3(b) shows two snapshots of the inclination profile as solid curves. The dotted curves represent the theoretical prediction, obtained from Equation (10) with the real-time Σ and ϵ, with ϵ measured using Equation (A15). The theory is seen to agree well with the simulations.

Panel (c) shows the measured and predicted γ profiles. The agreement between simulation and theory is adequate, although worse than for β. In linear theory, γ differs from zero due to the effect of ϵ, i.e. the non-Keplerianity, as seen in Equation (10) from the fact that the integral is purely real when $\epsilon = 0$. The mismatch in panel (c) is likely due to an insufficiently accurate evaluation of ϵ as the density evolves in time. But since γ has a subdominant effect on the warp, the agreement in panel (c) is not too concerning.

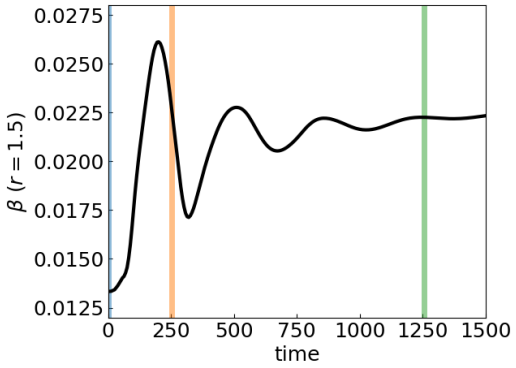


Figure 4. Time evolution of the disk inclination β at $r = 1.5$ for simulation C03, showing initially damped oscillations on a $t \sim 100$ timescale before settling into a quasi-steady state. The vertical colored lines correspond to the snapshot times shown in Figure 3.

Panel (d) shows the amplitude of the warp ψ (Equation 4), demonstrating good agreement with the linear theory prediction.

4. SLOSHING AND TORQUE

Having verified Equation (10) for the inclination profile, our goal in this section is to thoroughly explain the more fundamental linear equations (Equations (5)–(7)), in order to set the stage for the nonlinear theory presented next. We will also directly verify those equations with simulation C03. We discuss each of the three equations in turn.

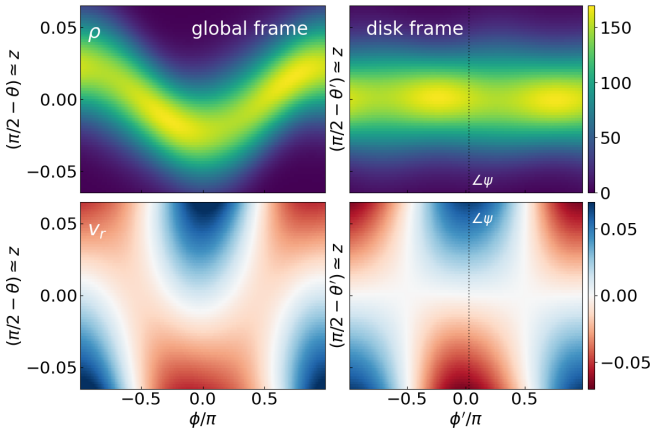


Figure 5. Cross-sections of density (ρ) and radial velocity (v_r), in order to illustrate sloshing. The figure shows cross-sections in simulation C03, at $r = 1.5$ and $t = 251.2$. **Left:** ρ and v_r in the global frame. **Right:** Same quantities but transformed into the local disk frame, with the disk midplane at $\theta' = \pi/2$. Specifically, we plot $\rho(\boldsymbol{r}')$ and $v_r(\boldsymbol{r}')$ with $\boldsymbol{r}' = R_y(\beta)R_z(\gamma)\boldsymbol{r}$, where $\boldsymbol{r}$ is the position vector in the simulation frame and R_y and R_z are the rotational matrices around the $\hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}}$ axis. The vertical dotted line marks the ϕ' angle where $\boldsymbol{\psi}$ is pointing.

Equation (5) is derived in Appendix A by showing that, at a given r, a warp $\boldsymbol{\psi}$ drives the velocity field

$$v_r = \frac{\psi}{4\alpha + 2i\epsilon} 2\Omega z e^{-i\phi} \ , \tag{11}$$

where $\psi \equiv \boldsymbol{\psi}\cdot(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})$ is the complexified warp, and the physical v_r is understood to be the real part of Equation (11).[3]

This warp-driven v_r is known as sloshing. Sloshing is shown in Figure 5, which plots a cross-section of the ρ and v_r fields at fixed $r = 1.5$ in simulation C03. In the left panel, these fields are plotted versus simulation-frame angles (θ and ϕ). The finite inclination at $r = 1.5$ makes those fields appear distorted. We remove the distortion in the right panels by rotating to a coordinate system aligned with the local $\hat{\boldsymbol{l}}$, denoted with primes (ϕ', θ'), as detailed in the caption. The density in this

[3] The cylindrical coordinates $\{z, \phi\}$ in Equation (12) are aligned with the unperturbed (unwarped) disk, as are the unit vectors $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$ used for complexifying $\boldsymbol{\psi}$. Although the spatial coordinates in Equation (11) are cylindrical, it is better to work with v_r rather than v_R, i.e., spherical rather than cylindrical radial velocity. That is because an unwarped disk has $v_r = 0$ (no sloshing), whereas it has $v_R \neq 0$ if it has finite inclination. Relatedly, v_r is unaffected by rotations of the coordinate system (or more precisely, it transforms as a scalar), whereas v_R mixes with v_ϕ and v_z. Finally, we note that in this paper, we do not adopt the common convention that unbolded vector symbols represent magnitudes. Thus, our ψ and $U_\perp$, for example, are complex valued.

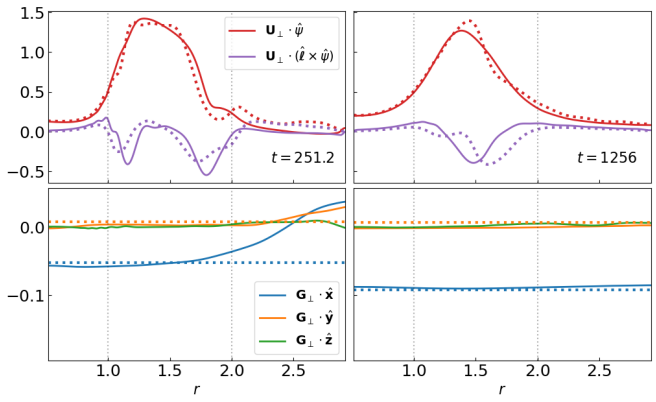


Figure 6. Sloshing vector $\boldsymbol{U}_\perp$ and internal torque $\boldsymbol{G}_\perp$ in simulation C03. **Top:** Components of the sloshing vector $\boldsymbol{U}_\perp$ (calculated as $\boldsymbol{U}_\perp = -\boldsymbol{G}_\perp/\sigma$), projected onto the $\hat{\boldsymbol{\psi}} \equiv \boldsymbol{\psi}/|\boldsymbol{\psi}|$ and $\hat{\boldsymbol{l}}\times\hat{\boldsymbol{\psi}}$ directions. **Bottom:** Cartesian components of $\boldsymbol{G}_\perp$ (measured by Equation 14) in the simulation frame. In all panels, the dotted lines represent linear theory predictions based on real-time Σ and ϵ profiles.

rotated frame (top-right panel) is nearly the same as an unperturbed flat disk (except for a faint hint of an $m = 2$ bouncing pattern, which is a nonlinear effect; see Section 6.3). And v_r exhibits sloshing (lower-right panel): fluid at a fixed height above the local midplane (fixed θ') sloshes in and then out as it orbits the star (i.e., as ϕ' increases), consistent with Equation (12). The sloshing is equal and opposite below the midplane.

The physical mechanism governing Equation (11) is described in, e.g., Figure 1 of Dullemond et al. (2022), and briefly recapped here. Consider two neighboring circular rings of fluid, an inner and an outer, each of which has a vertical extension given by the scale height. A warp means that the inner ring's midplane is tilted relative to the outer's. And since pressure is highest at the midplane, the inner will exert an outward force from its tilted midplane, with a spatial pattern given by Equation (11), in coordinates aligned with the outer ring. The small denominator in that equation appears because of the near-resonance between the epicyclic and orbital frequencies (e.g., Papaloizou & Pringle 1983).

We convert from Equation (11) to Equation (5) by parameterizing the amplitude of the v_r field at each r with the complex number $U_\perp$, defined via

$$v_r = 2U_\perp \Omega z e^{-i\phi} \ . \tag{12}$$

The coefficient of 2 has been chosen to make our $U_\perp = Q_4\psi$, where Q_4 is the commonly-used parameter defined in Ogilvie & Latter (2013a). In terms of this parameterization, Equation (11) becomes

$$U_\perp = \frac{\psi}{4\alpha + 2i\epsilon} \ . \tag{13}$$

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It is equivalent to the vectorial Equation (5), after converting from complex variables to vectors in the usual way, i.e., $\boldsymbol{U}_\perp = (\hat{\boldsymbol{x}}\mathrm{Re} + \hat{\boldsymbol{y}}\mathrm{Im})\{U_\perp\}$, and similarly for $\boldsymbol{\psi}$. Whereas Equation (11) is useful for appreciating the small-scale dynamics, Equation (5)) is more useful for connecting to the disk's global structure. Since the vector $\boldsymbol{U}_\perp$ will play an important role, we highlight its interpretation. Its magnitude is the same as that of $U_\perp$ in Equation (12), and thus quantifies the amplitude of the sloshing field. And its direction lies in the plane of the disk (perpendicular to $\hat{\boldsymbol{l}}$), and more specifically pointing in the direction where v_r is most positive above the midplane. In the lower-right panel of Figure 5, the direction is seen to be at a value of ϕ'/π that is slightly positive. Equation (13) predicts that the direction is the same as that of ψ (since here $|\epsilon| \ll \alpha$), as is seen to be the case in the figure.

Equation (6) relates $\boldsymbol{U}_\perp$ to the "perpendicular advective torque," defined as

$$\boldsymbol{G}_\perp(r) \equiv \left[\int \rho v_r \left(\boldsymbol{r}\times\boldsymbol{v}\right) r^2 d\Omega\right]_\perp , \tag{14}$$

where the $\perp$ subscript on the right-hand side stands for the components of the bracketed vector that are perpendicular to $\hat{\boldsymbol{l}}$, i.e., $[\boldsymbol{V}]_\perp \equiv [\boldsymbol{V}] - \hat{\boldsymbol{l}}(\hat{\boldsymbol{l}}\cdot[\boldsymbol{V}])$, for a generic vector $\boldsymbol{V}$. The bracketed vector is the advective flux of angular momentum across a shell of radius r.[4] We derive Equation (6), $\boldsymbol{G}_\perp = -\sigma\boldsymbol{U}_\perp$, in Appendix B, by inserting the sloshing field of Equation (12) into the definition of $\boldsymbol{G}_\perp$ in Equation (14), and working to linear order.

In order to verify the sloshing equation from simulations, one can extract the complex-valued $U_\perp$ from v_r in the code by taking the Fourier-Hermite transform of Equation (12), in the frame of the local $\hat{\boldsymbol{l}}$. However, we instead work everywhere with vectors rather than complex amplitudes. Specifically, we "measure" $\boldsymbol{U}_\perp$ in the simulation by measuring $\boldsymbol{G}_\perp$ via its definition (Equation 14), and then set $\boldsymbol{U}_\perp = -\boldsymbol{G}_\perp/\sigma$. We then compare that with the prediction from the sloshing equation,

$$\boldsymbol{U}_\perp = \frac{\alpha}{4\alpha^2+\epsilon^2}\boldsymbol{\psi} \;-\; \frac{\epsilon}{2\left(4\alpha^2+\epsilon^2\right)}\hat{\boldsymbol{l}}\times\boldsymbol{\psi} \tag{15}$$

$$\approx \frac{1}{4\alpha}\boldsymbol{\psi}, \quad \text{for } |\epsilon| \ll \alpha \tag{16}$$

[4] The true total torque is defined in Equation (C72). It differs from the square-bracketed expression in Equation (14) in two ways. First, the total torque also includes an accretional contribution, which is parallel to $\hat{\boldsymbol{l}}$, and thus is eliminated by the $\perp$-projection. And second, the total torque has a viscous contribution, which we exclude from our definition of $\boldsymbol{G}_\perp$ because its contribution is small.

which follows from inverting Equation (5), or by inspection from Equation (13). We verify Equation (15) in the top panels of Figure 6, where we show measurements of the components of $\boldsymbol{U}_\perp$ at two times in simulation C03. These are seen to agree with the prediction from Equation (15).

Equation (7) implies that $\boldsymbol{G}_\perp$ is independent of r, as must be true or a steady disk, from conservation of angular momentum. We verify this in the bottom panels of Figure 6, where we show the Cartesian components of $\boldsymbol{G}_\perp$, in the code's global coordinate system. It is seen that $\boldsymbol{G}_\perp$ in simulation C03 (solid) is indeed nearly constant. The discrepancy at $r \gtrsim 2$ in the early snapshot (top-left panel) indicates that the outer disk has not yet fully relaxed into a warp steady state. The dotted lines in the top panels show the prediction from linear theory, which is obtained from evaluating the constant in Equation (10), as described below that equation. The agreement is very good. The $\hat{\boldsymbol{z}}$ component of $\boldsymbol{G}_\perp$, which should be zero in linear theory, is indeed very small, as shown by the green curves in the bottom panel.

5. NONLINEAR THEORY

5.1. *Nonlinear Equations*

The linear theory shows that a warped disk is governed by two coupled processes: the local warp $\boldsymbol{\psi}$ drives a sloshing motion $\boldsymbol{U}_\perp$, and this sloshing generates a internal torque $\boldsymbol{G}_\perp = -\sigma\boldsymbol{U}$ that uphold the global inclination profile.

In the nonlinear regime, this picture remains qualitatively true, but the details and the governing equations are different. Equation (7) becomes impossible: because the $\perp$ components are not always in the $\hat{\boldsymbol{x}}$-$\hat{\boldsymbol{y}}$ plane, the direction of $\boldsymbol{G}_\perp$ must change with r. We instead only consider the magnitude $|\boldsymbol{G}_\perp|$ to be a constant, i.e.,

$$\partial_r|\boldsymbol{G}_\perp| \simeq 0. \tag{17}$$

This equation has a similar interpretation as Equation (7), except for being insensitive to the direction of $\boldsymbol{G}_\perp$. It turns out that Equation (**??**) is adequately accurate for a broad range of warp amplitudes, as we will see in the simulations (Section 6). We derive Equation (**??**) in Appendix C, showing that it is good approximation for the torque profile as long as the mass accretion rate of the disk is low.

The nonlinear relation between the internal torque and the warp can be written as

$$\boldsymbol{G}_\perp = -\sigma\left(Q_2\boldsymbol{\psi} + Q_3\hat{\boldsymbol{l}}\times\boldsymbol{\psi}\right), \tag{18}$$

which is equivalent to

$$\boldsymbol{U}_\perp = Q_2\boldsymbol{\psi} + Q_3\hat{\boldsymbol{l}}\times\boldsymbol{\psi}, \tag{19}$$

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where Q_2 and Q_3 are functions of $|\boldsymbol{\psi}|$, α, and ϵ. Equation (18) is a generalization of Equation (??): the $\hat{\boldsymbol{z}}$ is replaced with $\hat{\boldsymbol{l}}$ so that $\boldsymbol{G}_\perp$ and $\boldsymbol{U}_\perp$ are always perpendicular to $\hat{\boldsymbol{l}}$; Q_2 and Q_3 can change with $|\boldsymbol{\psi}|$, making Equations (18) and (19) nonlinear. The theory for how to calculate the nonlinear Q_2 and Q_3 has been worked out in Ogilvie (1999) and Ogilvie & Latter (2013a).

5.2. *Nonlinear Solutions*

The nonlinear steady-state structure is determined by a simple "demand-and-supply" relation: the internal torque $\boldsymbol{G}_\perp$ required by Equation (??) must be equal to what the warp can generated in Equation (18). The resulting solutions can, to a large extent, be understood graphically.

Figure 7 illustrates this process. We assume a σ profile as shown in panel (a), corresponding to the disk profile in Figure 3. Equation ?? implies $|\boldsymbol{U}_\perp| \propto \text{const}/\sigma$. Therefore, in panel (b) we plot three curves of const/σ that correspond to different values of $|\boldsymbol{G}_\perp|$. Those curves represent possible solutions for the global $|\boldsymbol{U}_\perp|$ profiles, which can be thought as the "demand" of $|\boldsymbol{U}_\perp|$ at each r.

Next, given $\boldsymbol{U}_\perp$ one may determine $\boldsymbol{\psi}$ by solving Equation (19). This can be done with the assistance of panel (c), where we plots Equation (19) for a range of $|\boldsymbol{\psi}|$. The coefficients Q_2 and Q_3 are calculated using the local shearing-box method introduced in Ogilvie & Latter (2013a), with $\alpha = 0.019$ and $\epsilon = 0$ for simplicity.[5] The calculation shows that, while $|\boldsymbol{U}_\perp|$ grows linearly with $|\boldsymbol{\psi}|$ when $|\boldsymbol{\psi}|$ is small, the relation becomes nonlinear and eventually saturates as $|\boldsymbol{\psi}|$ becomes large. The sloshing amplitude reaches a maximum $\simeq 4$ at a critical warp $|\boldsymbol{\psi}|_{\rm crit} \simeq 0.3$. At even larger warp amplitude, the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ relation enters a declining branch where further increases in $|\boldsymbol{\psi}|$ actually reduces the sloshing amplitude. This curve is the "supply" of $|\boldsymbol{U}_\perp|$ at each amplitude $|\boldsymbol{\psi}|$.

To match the $|\boldsymbol{U}_\perp|$ demand in panel (b) at all r with the supply given by panel (c), the disks must adopt the warp profiles $|\boldsymbol{\psi}|$ as shown in panel (d). In general, one must solve the two vectorial components of Equation (19) simultaneously. But for the purposes of presenting a simplified solution, we consider only the amplitude of that equation. Here we only consider the increasing branching of the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ in panel (c), because the decreasing branch is unstable as shown in Section 5.3.

Finally, the steady-state inclination profile $\beta(r)$ can be inferred by integrating $\boldsymbol{\psi}$, as shown in panel (e). Repeating this process for a range of $|\boldsymbol{G}_\perp|$, we get a corresponding range of outer inclinations $\beta_{\rm out}$. Panel (f) shows the maximum values of $|\boldsymbol{U}_\perp|$ and $|\boldsymbol{\psi}|$ achieved by these steady-state solutions, as functions of $\beta_{\rm out}$.

5.3. *Stability and Maximum $\beta_{\rm out}$*

We now look at the green curves (and dots) in Figure 7, where $\beta_{\rm out} \simeq 0.3$. They represent a critically warped disk where the $|\boldsymbol{\psi}|$ profile reaches $|\boldsymbol{\psi}|_{\rm crit}$ and the sloshing amplitude fully saturates. As seen in panel (c), this solution spans the entire increasing branch of the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ curve, including the regime where $|\boldsymbol{U}_\perp|$ is nearly flat. Consequently, the disk exhibits a strongly concentrated $|\boldsymbol{\psi}|$ profile with a sharp peak $|\boldsymbol{\psi}|$ at around $r = 1.2$ (panel d), corresponding to the weakest point.

What happens if we further increase $\beta_{\rm out}$? There would be no solutions unless the disk enters the decreasing branch of panel (c), where $|\boldsymbol{\psi}| > |\boldsymbol{\psi}|_{\rm crit}$. However, this decreasing branch is susceptible to a dynamical instability. Consider a disk that is slightly over-warped from the steady-state structure. If $|\boldsymbol{U}_\perp|$ increases with $|\boldsymbol{\psi}|$ (i.e., $\partial_{|\boldsymbol{\psi}|}|\boldsymbol{U}_\perp| > 0$), the over-warp would generate a stronger sloshing (i.e., a stronger internal torque) that acts to restore the equilibrium. Conversely, on the declining branch (i.e., $\partial_{|\boldsymbol{\psi}|}|\boldsymbol{U}_\perp| < 0$), the small over-warping would weaken the sloshing (i.e., weakens the restoring torque), leading to a runaway growth of the warp amplitude. We show this instability mathematically in Appendix C.3 (also see Doğan et al. 2018).

As a result, no continuous steady-state solution can be maintained if the required warp amplitude exceeds $|\boldsymbol{\psi}|_{\rm crit}$. This instability imposes a physical limit on the maximum stable $\beta_{\rm out}$. If $\beta_{\rm out}$ continues to increase, smooth steady-state warp profiles must then fail, leading to the so-called disk breaking.

6. NONLINEAR STEADY-STATE SIMULATIONS

We now examine the nonlinear regime and test the steady-state theory using simulations C10 to C30 (Table 1), for which the outer boundary inclination $\beta_{\rm out}$ varies from 0.1 and 0.3. Our goals are threefold. First, we determine whether these disks can maintain steady and continuous warp. Second, we test whether their global torque profiles and local sloshing-warp relation agree with the theoretical predictions from Section 5 (i.e., Equations ?? and 19). Third, we use the detailed

[5] To obtain Q_2 and Q_3, we first solve Equations (74)–(78) in Ogilvie & Latter (2013a), which are coupled time-dependent equations for the fluid velocities and densities, with parameters $|\boldsymbol{\psi}|$, α, and ϵ; we find the periodic solutions to these equations with a shooting method. We then insert the solutions into their Equation (92), which gives $Q_4|\psi| = (Q_2 + iQ_3)|\psi|$. That quantity is equivalent to our $Q_2\boldsymbol{\psi} + Q_3\hat{\boldsymbol{l}} \times \boldsymbol{\psi}$, after accounting for the correspondence between the complex and vector representations (Appendix B).

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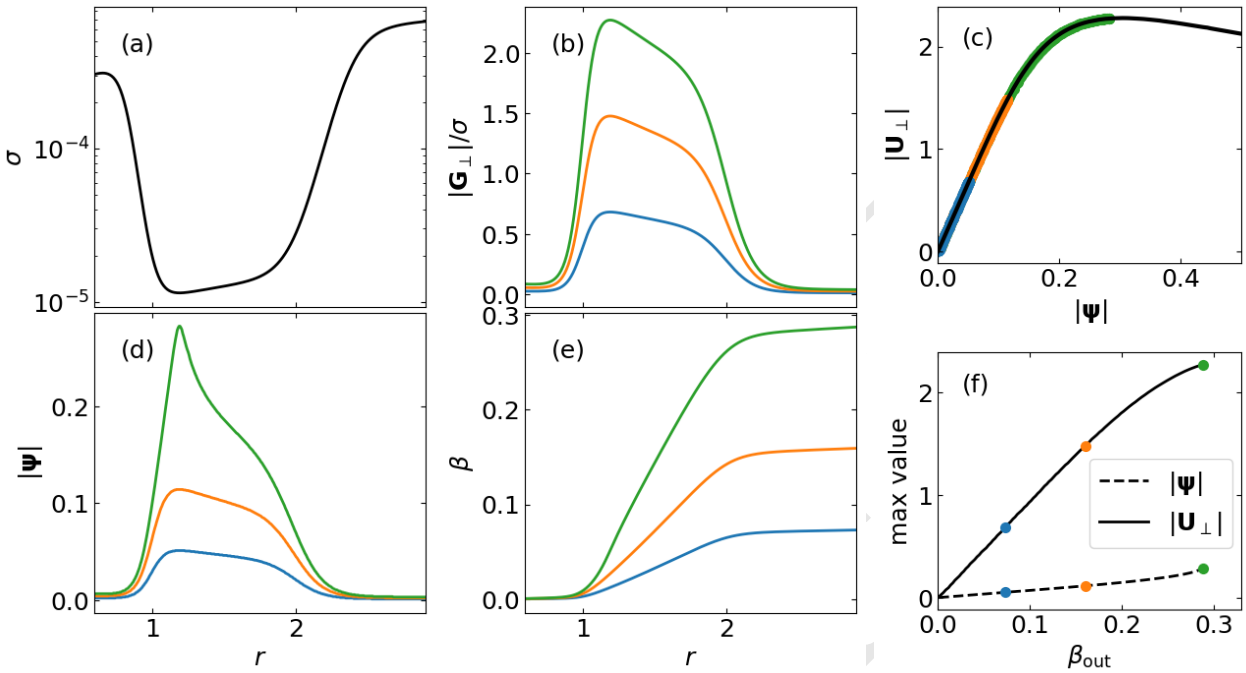


Figure 7. Example solutions to the nonlinear steady-state warp equations (**??**) and (18). **(a)** Fiducial radial profile of σ, given by Equation (8). **(b)** Global profiles for the sloshing amplitude $|\boldsymbol{U}_\perp| \equiv |\boldsymbol{G}_\perp|/\sigma$, based on Equation (**??**); the three colors represent solutions for different values of $|\boldsymbol{G}_\perp| = $ const, same for the other panels. **(c)** Local nonlinear relation between the sloshing $|\boldsymbol{U}_\perp|$ and warp amplitude $|\psi|$ based on Ogilvie & Latter (2013a); the colored dots indicates the values of $|\boldsymbol{U}_\perp|$ from (b). **(d)** Resulting $|\psi|$ profiles obtained by matching the two $|\boldsymbol{U}_\perp|$'s in (b) and (c). **(e)** Corresponding inclination profiles β. **(f)** Maximum warp and sloshing amplitude, $|\psi|$ and $|\boldsymbol{U}_\perp|$, as functions of the outer inclination $\beta_{\rm out}$; the curve terminates at the critical $\beta_{\rm out}$ where $|\psi| = |\psi|_{\rm crit}$.

simulation results to examine the internal structure and behaviors of warped disks.

To facilitate convergence, we initialize the disks using linear solutions for the inclination profile (β, γ) and sloshing velocity (v_r). While these initial conditions are not solutions to the nonlinear warp equations, they allow the tilting profiles to settle quickly into nonlinear steady states.

6.1. *Nonlinear steady states*

Figure 8 shows the time evolution of Σ, β, and $|\psi|$ at $r = 1.5$ for simulations C10 to C30. In all cases, the initial transient damps on a timescale $t \sim 50$, after which the disk enters a quasi-steady state. Beyond that time, the subsequent evolution is much slower and is driven mainly by the gradual filling of the density gap. We therefore regard these disks as being in warp steady states at $t > 100$. The analysis in the remainder of this Section will focus on the snapshots at $t = 125.6$.

Figure 9 summarizes the structures of these steady states. As $\beta_{\rm out}$ increases, the overall warp amplitude $|\psi|$ grows, while remaining concentrated in the density gap. The linear theory, shown by the faint curves in the right panels, provides a reasonable approximation to the simulated β and $|\psi|$ profiles, except for the $\beta_{\rm out} = 0.3$ case (red). We will see in the following subsections that nonlinear effects are important in setting the detailed structure of these disks. For now, Figures 8 and 9 complete our first goal, showing that nonlinear disks can maintain continuous and steady warps.

6.2. *Internal Torque*

We now test Equations (**??**) and (19), the two nonlinear steady-state relations from Section 5.

Equation (**??**) says the in-plane torque magnitude $|\boldsymbol{G}_\perp|$ is approximately constant. To test it, we measure $\boldsymbol{G}_\perp$ using Equation (14). The top panel of Figure 10 shows $|\boldsymbol{G}_\perp|^2$ as solid lines. In the warped region $(1 < r < 2)$, $|\boldsymbol{G}_\perp|^2$ is nearly constant for the simulations with $\beta_{\rm out}$ up to 0.2, consistent with Equation (**??**).

The C30 run (red) shows a more noticeable radial gradient. As we mentioned in Section 5.1, the condition $\partial_r |\boldsymbol{G}_\perp| \simeq 0$ is only an approximation because it neglects the effects of the mass accretion rate $\dot{M}$ and the slow time evolution of $\hat{\boldsymbol{l}}$. Appendix C.2 shows that the

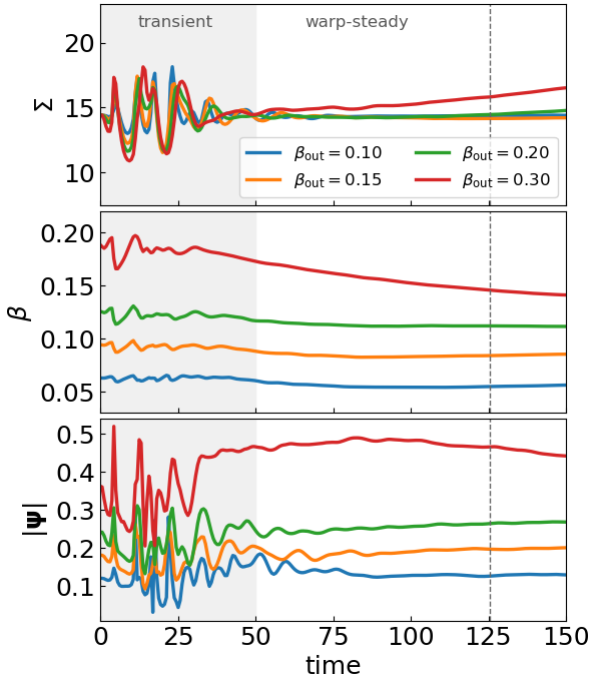


Figure 8. Time evolution of the disk density Σ, inclination β and warp amplitude $|\boldsymbol{\psi}|$ at $r = 1.5$ in simulations with $\beta_{\rm out}$ from 0.1 to 0.3 (C10 to C30 in Table 1). The vertical dashed lines mark $t = 125.6$, at which we analyze the simulation snapshots in Figures 9 to 13.

corrected expression is given by

$$\partial_r |\boldsymbol{G}_\perp|^2 \simeq -2\boldsymbol{G}_\perp \cdot \underbrace{\left(r\Omega\dot{M}\boldsymbol{\psi} + 2\pi\Sigma\Omega r^2 \partial_t \hat{\boldsymbol{l}}\right)}_{\partial_r \boldsymbol{G}_\perp} . \quad (20)$$

The first term in $\partial_r \boldsymbol{G}_\perp$ is the $\dot{M}$ correction. When mass flows across a warped region, its orbital plane must tilt as $\hat{\boldsymbol{l}}$ changes with radius (i.e., $\boldsymbol{\psi} = r\partial_r \hat{\boldsymbol{l}} \neq 0$). This change of orbital orientation requires a torque from the disk, which gives rise to the $r\Omega\dot{M}\boldsymbol{\psi}$ term in $\partial_r \boldsymbol{G}_\perp$. The second term accounts for the slow time evolution of $\hat{\boldsymbol{l}}$. Although $\partial_t \hat{\boldsymbol{l}}$ should be small in a warp steady state, it can become non-negligible when the background surface density evolves rapidly, which can occur when $\dot{M}$ is large. In simulation C30, we find that the $\dot{M}\boldsymbol{\psi}$ term is the primary source of $\partial_r \boldsymbol{G}_\perp$, while the $\partial_t \hat{\boldsymbol{l}}$ contribution is much smaller.

The bottom panel of Figure 10 compares both sides of Equation (20), using quantities measured directly from the simulations, and shows excellent agreement. The corrected torque profiles, $|\boldsymbol{G}_\perp|^2 + \int \boldsymbol{G}_\perp \cdot \partial_r \boldsymbol{G}_\perp dr$, shown as dotted curves in the top panel, are nearly spatially constant even in the C30 run.

We next demonstrate Equation (19) and determine the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ relation. In Figure 11, we measure $\boldsymbol{U}_\perp$ and $\boldsymbol{\psi}$ from the simulations and plot their values at $r = 1.2$, 1.3, ..., and 1.6, i.e. near the center of the density gap where ϵ is close to zero. The solid black line shows the nonlinear prediction of $|\boldsymbol{U}_\perp|$-vs-$|\boldsymbol{\psi}|$ from the local shearing-box model in Ogilvie & Latter (2013a) for $\alpha = 0.019$ and $\epsilon = 0$. For comparison, the dashed line is the linear prediction. The simulated results show an excellent agreement with the nonlinear prediction. Both theory and simulation show that the sloshing amplitude saturates at $|\boldsymbol{U}_\perp| \approx 4$ when $|\boldsymbol{\psi}| \sim 0.3$. At large $|\boldsymbol{\psi}|$, the $|\boldsymbol{U}_\perp|$ curve flattens and begins to decrease, foreshadowing the instability that triggers disk breaking (Section 7).

6.3. *Detailed Looks: ρ and v_r*

To understand how the nonlinear warp saturates, we now examine the internal fluid structure of the disks in more detail. Figure 12 shows the density and radial velocity fields near the center of the warped region, revealing how the disk departs from the linear regime once the warp becomes sufficiently large.

The left column of Figure 12 shows the density distribution in the local disk frame. In contrast to the weakly warped case, these nonlinear disks exhibit the so-called "bouncing" effect, characterized by the extreme vertical compression and expansion of the gas as it circulates along the $\hat{\boldsymbol{\phi}}$ direction, with a $m = 2$ azimuthal symmetry. This effect becomes increasingly dramatic toward larger warp amplitudes. Physically, these bouncing patterns correspond to the $m = 2$ breathing mode of the disk, resonantly excited by a periodic vertical driving force. This forcing arises from the second-order coupling between the warping and sloshing motions, occurring at twice the orbital frequency (Kaaz et al. 2025).

To quantify this bouncing behavior, we measure the local scale height $\tilde{H}$, defined as the standard deviation of the vertical density distribution:

$$\tilde{H}(r, \phi') = \left[\frac{1}{\Sigma}\int_0^\pi \rho\left(\theta' - \frac{\pi}{2}\right)^2 r^3 d\theta'\right]^{1/2}, \quad (21)$$

where $\theta' = \pi/2$ is the polar angle of the disk midplane. The resulting $\tilde{H}$ is shown by the solid white curves in the left column of Figure 12. In the most extreme case (simulation C30 with $\beta_{\rm out} = 0.3$), $\tilde{H}$ at $r = 1.5$ varies by nearly an order of magnitude over azimuth, from a minimum of 0.011 to a maximum of 0.097, compared with the unperturbed value $H = 0.037$. We compare these measurements to the nonlinear shearing-box theory of Ogilvie & Latter (2013a). We calculate the theoretical prediction using their Equations (74) to (78), with $\alpha = 0.019$, $\epsilon = 0$, and the simulated $|\boldsymbol{\psi}|$ values. The predicted values are shown as the dotted lines in Figure 12, which match the simulation results with remarkable accuracy. We also measure the $m = 2$ Fourier

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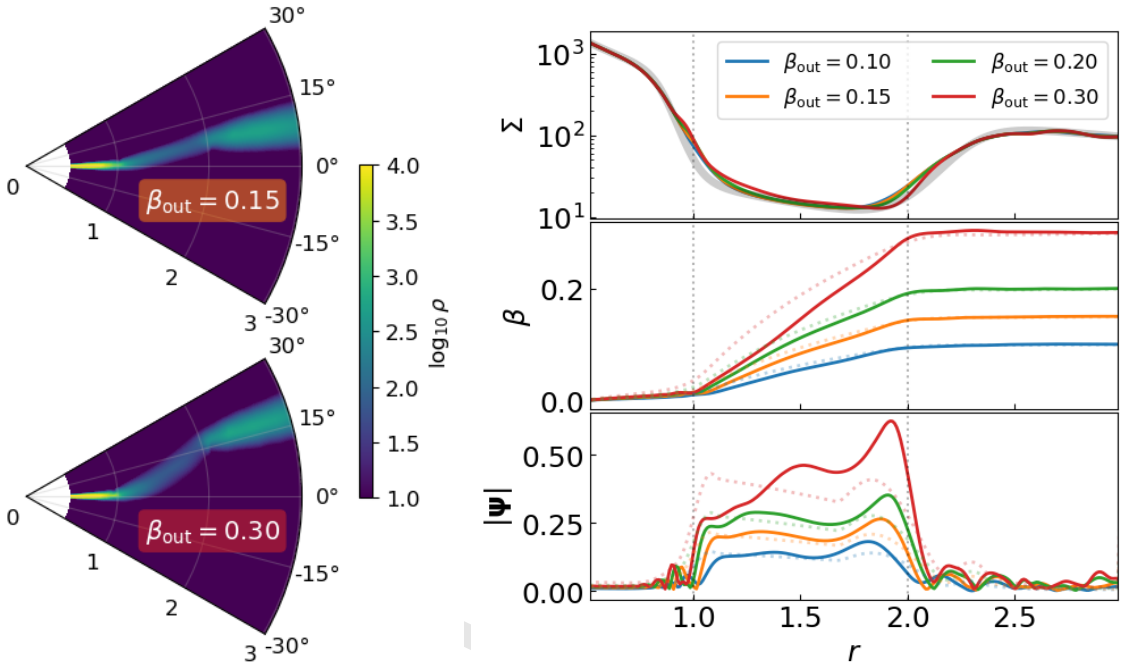


Figure 9. Snapshots of simulations with $\beta_{\rm out} = 0.1$ to 0.3 (C10 to C30) at $t = 125.6$, which all show steady and continuous warps. **Left:** Two side views of the disk volume density ρ ($\phi = 0$). **Right**: Surface density Σ, inclination β, and warp amplitude $|\psi|$ profiles. The solid lines are the measurements from simulations. The faint gray line in the top panel shows the initial density, while the colored faint lines in the other two panels show the linear theory predictions.

amplitude of $\tilde{H}$ at a few different radii, and plot it as a function of $|\psi|$ in the top-right panel of Figure 12 (same as how we measure the $|\boldsymbol{U}_\perp|$-$|\psi|$ relation in Figure 11). Comparing the simulations with the prediction from the nonlinear shearing-box theory, we find that the $m = 2$ amplitude of $\tilde{H}$ scales approximately as $|\psi|^2$ for $|\psi| \lesssim 0.3$. This quadratic scaling supports the interpretation that the bouncing mode is produced by a second-order nonlinear coupling. At larger $|\psi|$, however, the simulated amplitudes lie systematically below the theoretical prediction. This discrepancy is likely because the local shearing-box calculation keeps only the lowest-order Hermite term (i.e., $\rho(z')$ is strictly gaussian), whereas the simulations can contain contributions from higher-order vertical Hermite modes ($n > 1$) once the disk becomes strongly nonlinear.

To explain how this bouncing effect may play an important role in the saturation of the internal torque, we first turn to the radial velocity field v_r, shown in the middle column of Figure 12. In the weakly nonlinear regime, the v_r pattern is primarily dominated by the $m = 1$ azimuthal structure. As the warp becomes stronger, however, the v_r field develops clear higher-order azimuthal structures, with the strongest distortions occurring near the azimuths where the disk is most vertically compressed. This already suggests that the bouncing is potentially feeding back onto the sloshing flow, redistributing the original $m = 1$ sloshing motion into harmonics with $m > 1$.

To quantify the distortion of the v_r field, we calculate the Fourier-Hermite coefficients

$$U_{r,(m,n)} = \frac{1}{\pi} \int \frac{v_r}{r\Omega} \zeta e^{-\zeta^2/2} e^{i\phi'} d\zeta d\phi' \quad (m \geq 1), \quad (22)$$

where $\zeta = (\theta' - \pi/2)/H$. We evaluate these coefficients at $r = 1.2, 1.3, \ldots, 1.6$ for all simulations. The middle-right panel of Figure 12 shows these measurements as a function of the local $|\psi|$, analogous to Figure 11 and the top-right panel of Figure 12. The leading-order term $U_{r,(1,1)}$ (i.e., the only nonzero term in the linear theory) saturates as the warp increases. At the same time, the higher-order terms odd modes, especially $U_{r,(3,1)}$ and $U_{r,(5,1)}$, become prominent and take over in the growth. This transition occurs at roughly the same critical $|\psi|$ at which $|\boldsymbol{U}_\perp|$ saturates. The nonlinear shearing-box theory captures this trend qualitatively: it predicts the growth of the $m = 3$ component through the coupling between the $m = 1$ sloshing motion and the $m = 2$ bouncing mode.

The origin of these higher-order modes is potentially related to the $m = 2$ bouncing mode, as we mentioned earlier. The left and middle columns of Figure 12 show that v_r is largest at the azimuths where the disk is most vertically compressed. This phase correlation is consistent with the explanation that the $m = 3$ mode of v_r

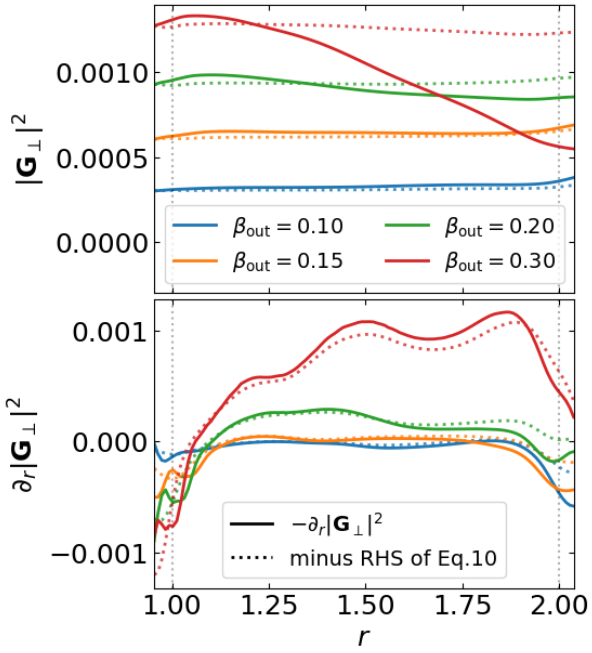


Figure 10. Diagnosis of the internal torque from the nonlinear simulations. We measure $\boldsymbol{G}_\perp$ directly using Equation (14) and compute $\partial_r \boldsymbol{G}_\perp$ from Equation 20. **Top:** Solid curves show the measurement of $|\boldsymbol{G}_\perp|^2$, while the dotted curves incorporates the $\dot{M}$ and $\partial_t \hat{\boldsymbol{l}}$ corrections by showing $|\boldsymbol{G}_\perp|^2 + \int \boldsymbol{G}_\perp \cdot \partial_r \boldsymbol{G}_\perp$. **Bottom:** Direct comparison between both sides of Equation (20), i.e. $\partial_r |\boldsymbol{G}_\perp|^2$ and the correction terms.

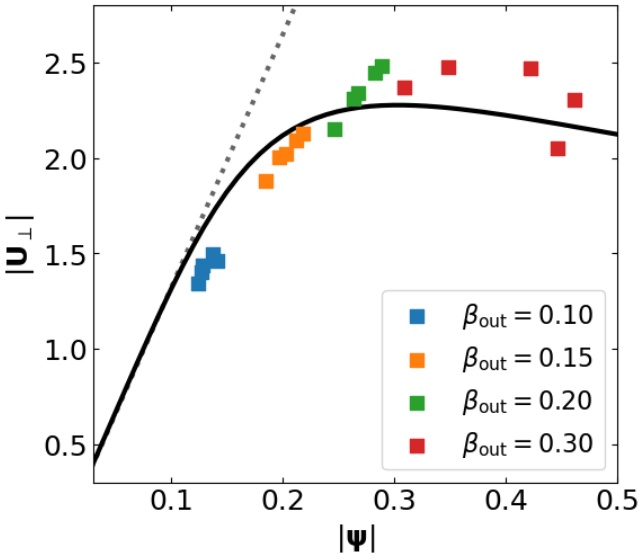


Figure 11. Relation between $|\boldsymbol{u}_r|$ and $|\boldsymbol{\psi}|$ in simulations with different $\beta_{\rm out}$ (i.e., C10 to C30 in Table 1). The black line shows the prediction for $\alpha = 0.0189$ and $\epsilon = 0$ based on the shearing-box model in Ogilvie & Latter (2013a), while the faint gray line represents the linear relation. The scattered squares show the values for $|\boldsymbol{u}_r|$ and $|\psi|$ measured from the simulations at $t = 125.8$. Each color represents one simulation with a different $\beta_{\rm out}$, while the squares with the same color correspond to the results measured at $r = 1.2$, 1.3, ..., 1.6.

arises from the coupling between the original $m = 1$ velocity field and the $m = 2$ bouncing mode in the density field. More generally, once the $m = 2$ bouncing becomes strong, it can couple odd-m velocity components into other odd-m components, which explains why we find $U_{r,(m,1)} \approx 0$ for even m.

This mode coupling provides a natural explanation for the saturation of the internal torque. From the integral definition of $\boldsymbol{G}_\perp$ in Equation (14), the factor $\boldsymbol{r} \times \boldsymbol{v}$ in the integrand has $m = 1$ azimuthal symmetry in the local disk frame. Therefore, only the $m = 1$ Fourier component of v_r can contribute coherently to $\boldsymbol{G}_\perp$; higher-order components such as $U_{r,(3,1)}$ and $U_{r,(5,1)}$ do not generate a net torque after azimuthal averaging. We verify this explicitly in the bottom-right panel of Figure 12, which shows that the $(m, n) = (1, 1)$ component alone reproduces essentially all of $|\boldsymbol{U}_\perp|$, while the contribution from the remaining modes is negligible.

We therefore argue that torque saturation is caused by the nonlinear redistribution of the radial motion from $m = 1$ into $m > 1$. At small warp amplitude, the growing warp mainly strengthens the coherent $m = 1$ sloshing, so $G_\perp$ and $U_\perp$ both increase linearly. At larger warp amplitude, the coupling to the $m = 2$ bouncing mode progressively transfers power into $m = 3$, $m = 5$, and other higher-order components. Because those modes do not contribute to the net internal torque, $G_\perp$ and $U_\perp$ saturate once the $m = 1$ sloshing is no longer the dominant growing component.

6.4. *Accretion Rate*

Another important consequence of nonlinear warping is the enhancement of the mass accretion rate. We measure the accretion rate $\dot{M}$ directly from our simulation using

$$\dot{M}(r) = \int v_r \rho r^2 d\Omega. \tag{23}$$

The resulting $\dot{M}$ within the density gap ($1 < r < 2$) is shown as colored curves in Figure 13, with the rate increasing steeply with warp amplitude.

This enhancement of the accretion rate is expected from the nonlinear theory, as a direct result of the warp-induced internal stresses. We show in Appendix C.1 that the accretion rate in a warped disk obeys

$$\dot{M}\partial_r\left(\Omega r^2\right) = -\partial_r\left(\boldsymbol{G}\cdot\hat{\boldsymbol{l}}\right) + \frac{1}{r}\boldsymbol{G}\cdot\boldsymbol{\psi}, \tag{24}$$

where $\boldsymbol{G}$ is the internal torque including the parallel components, which can be measured using Equa-

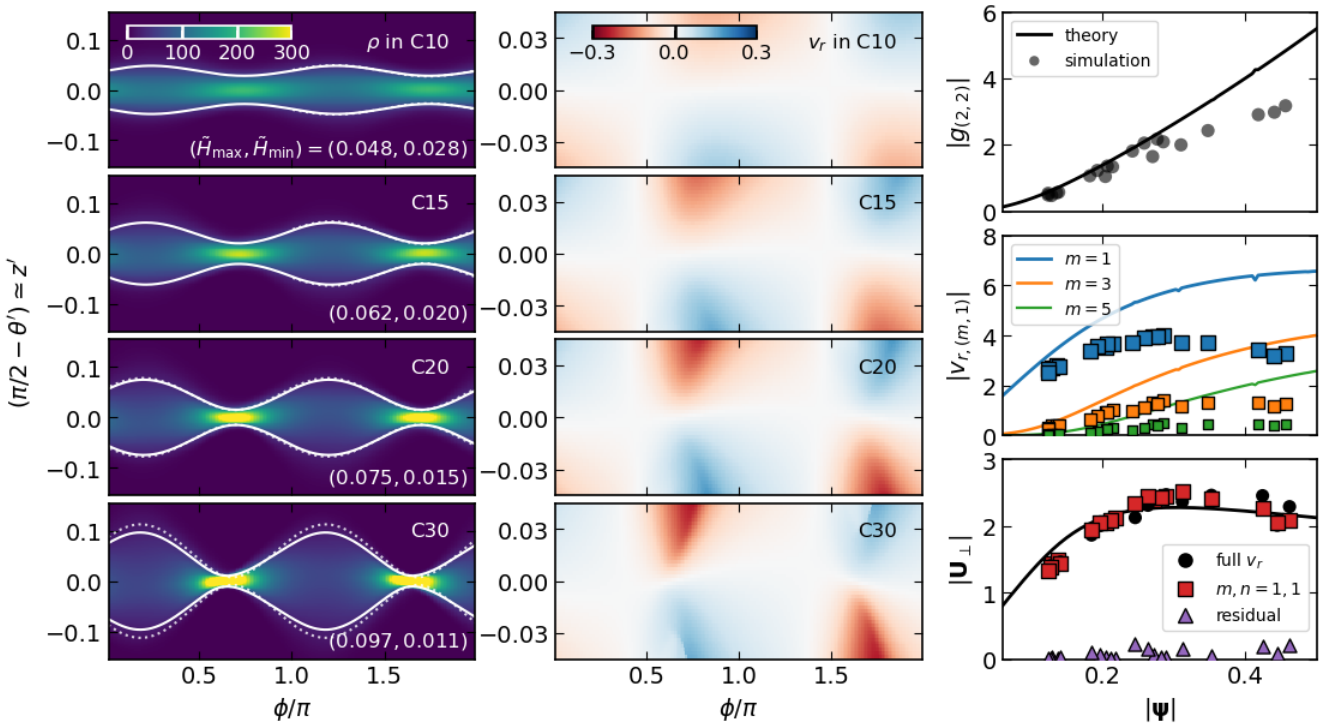


Figure 12. Azimuthal variation of the disk's density scale height and sloshing amplitude at $r = 1.5$ and $t = 125.6$ from simulations C10 to C30. **Left:** Density distribution in the disk frame (i.e., same as the top-right panel of Figure 5). The solid white lines trace the disk scale height measured using Equation (21), while the dotted lines are the predictions based on Ogilvie & Latter (2013a). **Right:** Solid colored lines show the local sloshing coefficient v_{r1} measured using Equation (??). The dotted colored lines are the predictions based on Ogilvie & Latter (2013a). The solid and the dotted black curves are the same as the colored ones, but only including the $m = 1$ and $m = 3$ Fourier components.

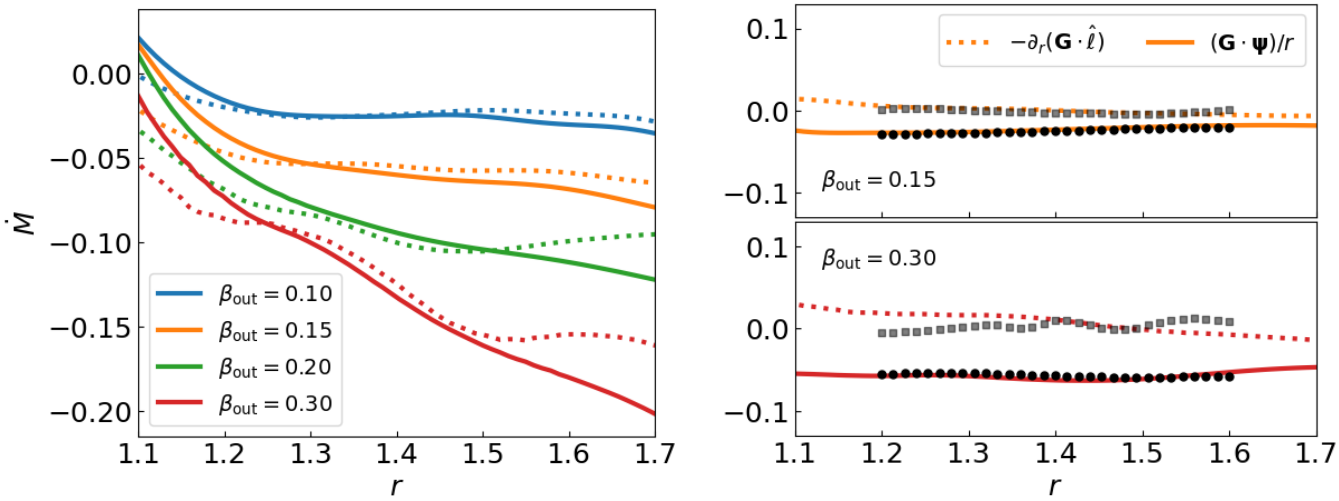


Figure 13. **Left:** Mass accretion rate $\dot{M}$ at $t = 125.6$ in the nonlinear steady-state simulations. The solid lines are measurement using Equation (23); the dotted curves show the prediction from Equation (24). **Right:** Decomposition of the right-hand side of Equation (24) for the $\beta_{out} = 0.15$ (top) and 0.30 (bottom) runs. The dotted colored curve shows $-\partial_r(\boldsymbol{G}\cdot\hat{l})$; the solid colored curve shows $(\boldsymbol{G}\cdot\boldsymbol{\psi})/r$. The gray boxes and black dots represent $\partial_r(\sigma Q_1)$ and $-\sigma Q_2|\psi|^2/r$, where Q_1 and Q_2 are calculated using the shearing-box model in Ogilvie & Latter (2013a) with the real-time local values of $|\psi|$, $\alpha = 0.019$, and $\epsilon = 0$.

tion (C75). This equation is equivalent to Equation (51) of Ogilvie (1999). We can further use the nonlinear

theory for $\boldsymbol{G}$ to predict the resulting $\dot{M}$. With Equation (18) implying $\boldsymbol{G}\cdot\boldsymbol{\psi} = -2\sigma Q_2|\boldsymbol{\psi}|^2$, by defining $Q_1 \equiv \boldsymbol{G}\cdot\hat{\boldsymbol{l}}/(-2\sigma)$, we get

$$\dot{M}\partial_r\left(\Omega r^2\right) = 2\partial_r\left(\sigma Q_1\right) - \frac{2}{r}\sigma Q_2|\boldsymbol{\psi}|^2. \tag{25}$$

Same as Q_2, the coefficient Q_1 can also be calculated using the shearing-box model from Ogilvie & Latter (2013a).

The first term on the right-hand sides of Equations (24) and (25) corresponds to the usual viscous accretion associated with the torque component parallel for small warps, while the second term is purely warp-driven. In the linear regime, $Q_1 \approx -3\alpha/2$ and $Q_2|\boldsymbol{\psi}|^2 = 0$. For sufficiently large warp, however both the Q_1 and $Q_2|\boldsymbol{\psi}|^2$ terms scale with $|\boldsymbol{\psi}|^2/\alpha$ as predicted by Equations (95) and (97) of Ogilvie & Latter (2013a). Consequently, the accretion rate is mostly contributed by the warp-induced stress rather the standard α viscosity when $|\boldsymbol{\psi}| > \alpha$.

To test the $\dot{M}$ prediction from Equation (24), we calculate the right-hand-side terms for each simulation using the measured $\boldsymbol{G}$, $\hat{\boldsymbol{l}}$ and $\boldsymbol{\psi}$ profile. The predicted $\dot{M}$ is plotted as dotted lines in the left panel of Figure 13, broadly consistent with the direct measurements.

The right panels show the two terms on the right-hand side of Equation (24), separately for the C15 and C30 runs. In both cases, the quantity $-\partial_r(\boldsymbol{G}\cdot\hat{l})$ is positive, contrary to the viscous accretion in flat disks. The second term $(\boldsymbol{G}\cdot\boldsymbol{\psi})/r$ is negative and has the larger magnitude, so their sum yields a net inward accretion flow. The contribution from $(\boldsymbol{G}\cdot\boldsymbol{\psi})/r$ becomes larger in the more strongly warped disk, consistent with the enhancement of $\dot{M}$ being primarily driven by the warp-induced internal torque.

The scattered points show the two terms on the right-hand side of Equation (25), namely $2\partial_r\left(\sigma Q_1\right)$ (gray boxes) and $-2\sigma Q_2|\boldsymbol{\psi}|^2/r$ (black dots). We evaluate Q_1 and Q_2 using the nonlinear shearing-box model in Ogilvie & Latter (2013a), adopting the real-time simulated values of $|\boldsymbol{\psi}|$, $\alpha = 0.019$ and $\epsilon = 0$, and then combine them with the simulated σ profiles. The shearing-box prediction for $-2\sigma Q_2|\boldsymbol{\psi}|^2$ agrees well with the simulated $\boldsymbol{G}\cdot\boldsymbol{\psi}$. The correspondence between $2\partial_r(\sigma Q_1)$ and $-\partial_r(\boldsymbol{G}\cdot\hat{\boldsymbol{l}})$ is less exact, which is possibly due to our assumption of $\epsilon = 0$ when evaluating Q_1 and the noises from taking numerical derivatives. Overall, the shear-box model accurately predicts the dominate contribution to $\dot{M}$, i.e. $-2\sigma Q_2|\boldsymbol{\psi}|^2 = (\boldsymbol{G}\cdot\boldsymbol{\psi})/r$. Therefore, it can be used as an estimate of accretion rate in nonlinear warped disks.

7. DISK BREAKING

Sections 5 and 6 have studied how disks maintain nonlinear and continuous warps. We now investigate the regime where such steady states fail. The analysis in Section 5.3 suggests that when the warp amplitude $|\boldsymbol{\psi}|$ exceeds a critical value, the internal torque can no longer provide a restoring response: $\partial_{|\psi|}|U|_\perp$ becomes negative, and the disk is expected to undergo a runaway amplification of the warp. Here we test that prediction with simulations.

7.1. *Breaking Simulation*

We begin with simulation C40, which is identical to the models in Section 6, except that the outer inclination is increased to $\beta_{\rm out} = 0.4$, the largest value in our suite. Unlike the other models (C10-C30) which relax into smooth continuous profiles, simulation C40 shows dramatic structural evolution.

The left panels of Figure 14 show side-view snapshots of the density field in C40. The disk begins from a continuous warp at $t = 0$, yet subsequently, the warped region narrows and steepens. As it evolves through $t \approx 62.8$ to 125.6, the inner and outer disk regions gradually detach, forming two disconnected planes separated at $r \simeq 1.5R_0$. At later times, each piece becomes approximately flat, with the inner and outer disks aligning toward $\beta_{\rm in} = 0$ and $\beta_{\rm out} = 0.4$, respectively.

The right-hand panels of Figure 14 show the same evolution in terms of the radial profiles of Σ, β, and $|\boldsymbol{\psi}|$. Initially, the inclination profile β is continuous, and the warp is distributed broadly across the gap, with $|\boldsymbol{\psi}| \sim 0.5$ overthe region $1 < r < 2$. As the disk evolves, β gradually steepens and the warp becomes strongly localized, and the disk eventually breaks at $r \simeq 1.5$. At breaking point, the warp amplitude reaches an maximum as large as $|\boldsymbol{\psi}| \sim 8$, while the surface density Σ at the same location drops by roughly an order of magnitude. The final state consists of two disconnected planes, with the inner and outer parts aligning to $\beta_{\rm in} = 0$ and $\beta_{\rm out} = 0.4$, separated by a sharp density dip at $r = 1.5$.

This result suggests that the $\beta_{\rm out} = 0.4$ disk cannot maintain a continuous warp. The disk instead breaks through a runaway growth of $|\boldsymbol{\psi}|$, which drives all warping to concentrate at one point, causing the disk to break. Importantly, this breaking process is a direct consequence of the internal hydrodynamics and completely spontaneous: no explicit external forces are applied except for the fixed boundary condition.

7.2. *Unbreaking Experiments*

To confirm that breaking reflects a genuine absence of continuous steady state, rather than merely an outcome of the initial condition, we perform an additional set of *unbreaking* simulations.

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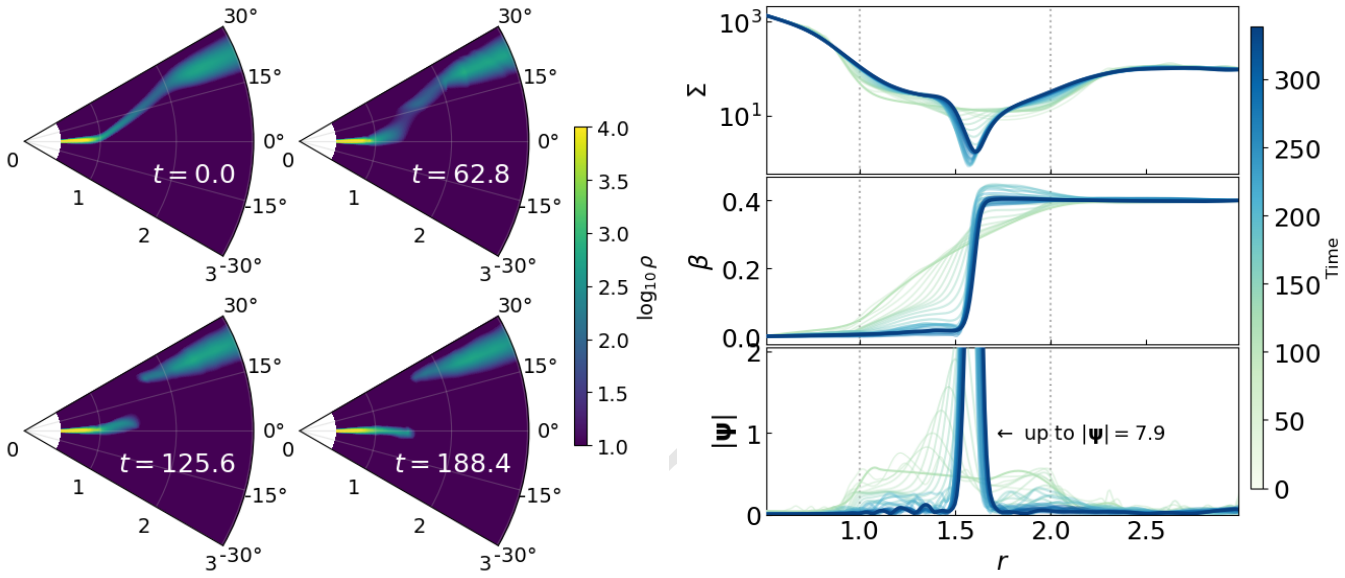


Figure 14. Results from the $\beta_{\rm out} = 0.4$ simulation (C40). **Left:** snapshots of the disk density maps. The disk begins with a continuous warp, but then the inner and outer disk regions gradually detach, eventually forming two disconnected planes. **Right:** time evolution of the disk density Σ, inclination angle β, and warp amplitude $|\boldsymbol{\psi}|$. This breaking process happens spontaneously without explicit perturbers.

In these runs, the disks start with nearly broken initial conditions. The initial background surface density profile is the same as before, but the initial inclination is imposed as a step function,

$$\beta(r) = \begin{cases} 0 & \text{if } r < 1.5, \\ \beta_{\rm out} & \text{if } r > 1.5. \end{cases} \tag{26}$$

No initial sloshing motion is added. We consider three values of the imposed outer inclination, $\beta_{\rm out} = 0.2$, 0.3, and 0.4. These runs will be referred to C20-unbreak, C30-unbreak, and C40-unbreak.

Figure 15 shows the evolutions in these simulations, clearly indicating a bifurcation in the outcomes. In the C20-unbreak and C30-unbreak runs, the inner and outer disk portions reconnect and settle toward smooth warped profiles. By the end of the simulations, the β and $|\boldsymbol{\psi}|$ profiles largely resemble the steady states found in Section 6.

In contrast, the C40-unbreak disk remains broken: despite some small evolutions, the β profile stays close to a step function, the warp $|\boldsymbol{\psi}|$ remains highly localized (nearly zero everywhere except at the breaking location), and the Σ profile gradually opens a narrow gap at the breaking location.

These simulations show that whether the disk ends up continuous or broken is controlled by the existence of a stable steady-state solution, which is determined by the imposed boundary misalignment in our setup. The reconnection (or *unbreaking*) in C20-unbreak and C30-unbreak suggest that the continuous steady states at $\beta_{\rm out} < 0.3$ are stable attractors, whereas no such state exists when $\beta_{\rm out} = 0.4$. Increasing the inclination from $\beta_{\rm out} = 0.3$ to 0.4 triggers a true transition: from being able to find and maintain a continuous warp, to a long-lived broken configuration.

7.3. *Comparison with Theory*

The breaking process shown simulation C40 is qualitatively consistent with the stability analysis in Section 5.3. There, we argue that a steady state solution becomes unstable once $\beta_{\rm out}$ is large enough, so that the disk is forced into the declining branch of the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ curve. We show in Section 5.3 and Appendix C.3 that, on the declining branch, local increase in warp amplitude weakens the internal torque and leads to a runaway growth of the warp. The evolution of simulation C40 matches this expectation closely: the warp localizes and steepens rapidly, and eventually becomes a break.

That said, quantitatively, the exact critical $\beta_{\rm out}$ for breaking differ in the simulations and the simplified nonlinear theory. Panels (c) and (f) of Figure 7 suggest that the breaking should occur for our fiducial disk as soon as $\beta_{\rm out}$ reaches 0.3, whereas the simulations show that the instability starts between 0.3 and 0.4. The results from simulation C30 highlights this discrepancy. As the red squares in Figure 11 show, even when $|\boldsymbol{U}_\perp|$ is fully saturated and the local $|\boldsymbol{\psi}|$ values are already over the predicted limit $|\boldsymbol{\psi}|_{\rm crit} \simeq 0.3$, the disk remains not broken.

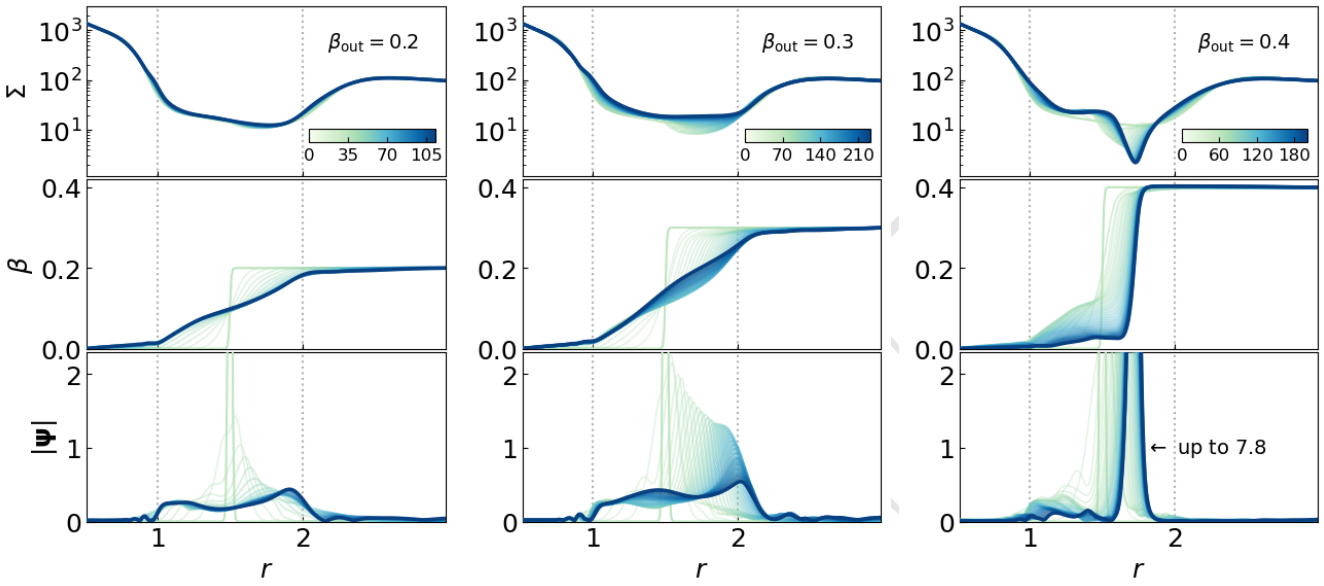


Figure 15. Surface density Σ, inclination profile β, and warp amplitude $|\psi|$ in three unbreaking experiments. In each column, the faint-to-dark colors show the time evolution from the nearly broken initial condition toward the final state. The C20-unbreak and C30-unbreak disks reconnect and relax toward continuous warped profiles, whereas the $\beta_{\rm out} = 0.4$ disk remains broken.

This discrepancy is at least partially due to the approximations in the simplified nonlinear theory in Section 5. There, steady-state solutions assume both $\partial_t \hat{\boldsymbol{l}} = 0$ and $\partial_r |\boldsymbol{G}_\perp| = 0$, but simulation C30 exhibits noticeable departures from both due to the enhanced $\dot{M}$ (Sections 6.2 and 6.4). Thus the critical $|\boldsymbol{\psi}|$ and $\beta_{\rm out}$ predicted by our theory is an idealized estimate; precise calculation requires incorporating the warp-induced $\dot{M}$ (and potentially other disk feedback). Although our theoretical prediction need not to exactly match the simulation results, it still provides a good estimation to the empirical value.

Another plausible explanation to why this disk does not break in C30 is shock dissipation. In a strongly warped, the bouncing motion becomes large, and some gas may encounter shocks when the compression becomes supersonic. Such shocks could in principle modify the internal fluid dynamics, potentially suppressing the growth of warp and changing the threshold for instability (Held & Ogilvie 2024; Kaaz et al. 2025). However, we do not find clear evidence of shocking in simulation C30. Moreover, recent work by Held & Ogilvie (2025) argues that shock may facilitate disk breaking in some cases, rather than suppressing it. We therefore argue that the shock dissipation is irrelevant to the C30 case.

We have repeated simulations C10 to C40 with both doubled and halved amounts of grid cells in the θ-direction, and we found the same critical $\beta_{\rm out}$ for breaking. Thus, the experimental breaking condition is numerically robust. That said, we note that post-breaking disks shows a slow outward drafting of the breaking location (see Figures 14 and 15), suggesting that the inner and outer parts of an unbroken disk may still be communicating. The rate of this outward drift appears faster in low-resolution runs. A follow-up study may be needed to investigate the long-term outcomes after disks break.

Overall, our breaking simulations are in good agreement with the theoretical picture. The breaking process is consistent with the runaway instability on the declining branch of the $|\boldsymbol{U}|$-$|\boldsymbol{\psi}|$ relation. The predicted critical $\beta_{\rm out}$ for breaking is reasonably accurate, despite a few simplifications.

7.4. *Breaking Conditions for Different α*

To extend this stability analysis beyond our $\alpha = 0.019$ disk model, we use the local shearing-box method of Ogilvie & Latter (2013a) to calculate the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ relation for other values of α, while keeping $\epsilon = 0$. Figure 16 shows how the result shifts systematically with α. The $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ relation becomes noticeably nonlinear once $|\boldsymbol{\psi}| \gtrsim \sqrt{\alpha}$. The sloshing amplitude reaches a maximum of $|\boldsymbol{U}_\perp| \simeq 0.6/\sqrt{\alpha}$, occurring at a critical warp amplitude $|\boldsymbol{\psi}|_{\rm crit} \simeq 2\sqrt{\alpha}$. This result implies that lower-viscosity disks are more vulnerable to breaking, because the maximum warp amplitude they can sustain before entering the unstable regime is given by $|\boldsymbol{\psi}|_{\rm crit} \simeq 2\sqrt{\alpha}$.

8. CONCLUSION

In this paper, we investigated warp steady states in nearly Keplerian disks, spanning the linear, nonlinear, and breaking regimes. Our analysis combines linear theory, nonlinear shearing-box model, and three-

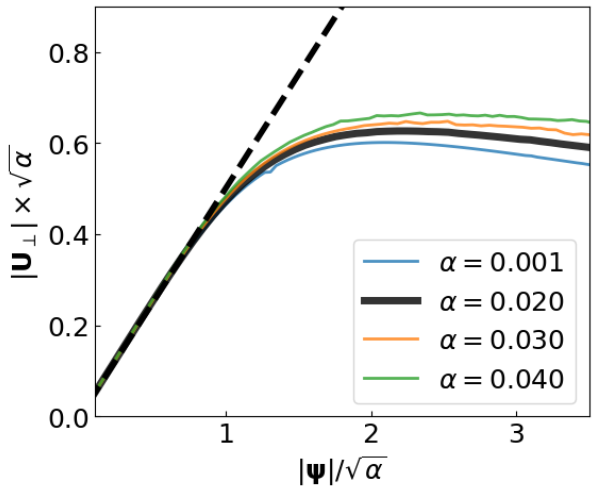


Figure 16. Steady-state relation between sloshing $|\boldsymbol{U}_\perp|$ and the warp amplitude $|\boldsymbol{\psi}|$ for the different α, with $\epsilon = 0$. The solid curves show the prediction from the shearing-box model (Ogilvie & Latter 2013a), while the dashed line is the linear theory (based on Equation 19). After rescaling both axes by $\sqrt{\alpha}$, the nonlinear curves for different α approximately becomes a single relation.

dimensional hydrodynamic simulations. Rather than introducing an explicit perturber or external torque, we simulate disks with inner and outer boundaries being fixed at different inclinations. This model provides a clean setup for imposing a (steady) warp to a disk, while remaining agnostic about the physical origin of the misalignment. We thus obtain a simple but comprehensive and predictive picture for the hydrodynamics of warped disks, illustrating how they sustain coherent warps or progress to breaking. Our main results are as follows.

For weak warps (Sections 3 and 3.3), the simulated disk (C03) relaxes rapidly into a warp steady state that is well described by the linear theory. The steady-state disk is found to bend the most at its weakest point (i.e. where the scaled density σ is the smallest), have a constant internal torque profile $\boldsymbol{G}_\perp$, and show a sloshing motion pattern that is azimuthally sinusoidal and vertically linear; all these results agree closely with the linear prediction. Although the background surface density evolves on the viscous timescale, the warp profile quickly adjusts and remains close to the instantaneous steady-state solution. This implies that, in protoplanetary disks, observed warps are likely to be in long-live steady states, rather than wave-like transients.

For moderately warped disks, the steady state enters a nonlinear regime (Sections 5 and 6, C10 to C30). The local relation between the internal torque $\boldsymbol{G}_\perp$ and the warp $\boldsymbol{\psi}$ departs from the linear theory, while the sloshing amplitude $\boldsymbol{U}_\perp$ begins to saturate (at $\boldsymbol{\psi} \simeq 2\sqrt{\alpha}$). At the same time, the disk develops several distinctive nonlinear features, including strong vertical "bouncing" motions (large azimuthal variations of density scale height) and substantially enhanced mass accretion rate. These effects are all measured from the simulations and show good quantitative agreement with the theoretical predictions based on the nonlinear shearing-box model of Ogilvie & Latter (2013a).

Sufficiently strong warps are unstable and can break the disk. In our simulation C40, the $\beta_{\rm out} = 0.4$ disk undergoes a runaway instability such that the warp amplitude $|\boldsymbol{\psi}|$ grows rapidly and becomes highly localized, eventually splitting the disk into two misaligned planes separated by a deep density dip. This breaking process occurs spontaneously due to internal hydrodynamics, without requiring any explicit external forcing besides the fixed boundary inclinations. We further perform a suite of unbreaking experiments (C20- to C40-unbreak), where we find that disks with nearly broken initial condition can reconnect and relax back to continuously warped structures when $\beta_{\rm out} \leq 0.3$, but remain broken when $\beta_{\rm out} = 0.4$. These results suggest that our disk breaking is not merely an artificial transient due to the initial condition, but rather related to the existence of a stable and continuous warp steady state. Such state exist when the mutual inclination between the boundaries is sufficiently small, but can vanish if otherwise.

We present a simplified nonlinear theory about steady warps and breaking in Section 5. The theory can be combined with the local model of Ogilvie & Latter (2013a) to solve for the global warp profile of a disk, but more importantly it highlights the underlying physics. For weakly and moderately warped disks, continuous and steady structures are maintained by two coupled processes: (i) the warp drives sloshing through local hydrodynamics, and (ii) the sloshing generates a nearly constant internal torque that upholds the global inclination profile. For strongly warped disks, our theory suggest a disk is locally unstable if the $|\boldsymbol{U}_\perp|$ and $|\boldsymbol{\psi}|$ are anti-correlated, so the nonlinear saturation of internal torque naturally leads to the onset of disk breaking. Using the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ relation calculated based on Ogilvie & Latter (2013a), the theory estimates that the critical $\beta_{\rm out}$ for breaking is 0.3 for the $\alpha = 0.019$ disk in our simulations, which is reasonably close to imperial value $\beta_{\rm out} = 0.4$. We also show that the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ relation scales with $\sqrt{\alpha}$ scaling in nearly-Keplerian disks, with the critical warp amplitude for disk breaking being $|\boldsymbol{\psi}|_{\rm crit} \simeq 2\sqrt{\alpha}$.

That said, there are several potential directions for future work. One is to use our model to investigate the long-term evolution of broken disks. For example, some

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previous simulations have shown disks undergoing multiple episodes of breaking and reconnecting (e.g., Deng & Ogilvie 2022), but the physical mechanism behind such behavior remains unclear. Another important extension is to generalize the breaking condition by incorporating additional physics, such as magnetic fields and turbulence, especially given that our results depend sensitively on α. Finally, the nonlinear effects analyzed in this paper, such as bouncing and the enhanced accretion rate, may produce observable signatures. Modeling those signatures in a suite of controlled experiment would not only help future warp detections, but also turn warps into a more powerful probe of disk physics.

ACKNOWLEDGMENTS

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APPENDIX

A. DERIVATION OF THE LINEAR THEORY: COMPLEX EQUATIONS

Here we derive the linearized equations of motion for a warp in a globally isothermal protoplanetary disk. The disk is governed by the momentum and continuity equations,

$$(\partial_t + \boldsymbol{v}_{\rm tot}\cdot\boldsymbol{\nabla})\,\boldsymbol{v}_{\rm tot} = -\boldsymbol{\nabla}\Phi - c_s^2\boldsymbol{\nabla}\lambda_{\rm tot}, \tag{A1}$$

$$(\partial_t + \boldsymbol{v}_{\rm tot}\cdot\boldsymbol{\nabla})\,\lambda_{\rm tot} = -\boldsymbol{\nabla}\cdot\boldsymbol{v}_{\rm tot}, \tag{A2}$$

where the subscript "tot" denotes the total value (background + perturbation), $\boldsymbol{v}_{\rm tot}$ is the velocity vector,

$$\lambda_{\rm tot} = \ln\rho_{\rm tot} \tag{A3}$$

is the logarithm of the density, c_s is the globally constant sound speed, and Φ is the gravitational potential of the central star. We temporarily ignore viscosity.

We shall decompose

$$\lambda_{\rm tot} = \lambda + \lambda' \tag{A4}$$

$$\boldsymbol{v}_{\rm tot} = \boldsymbol{v} + \boldsymbol{v}' \tag{A5}$$

where unsubscripted λ and $\boldsymbol{v}$ denote the background, and primed quantities denote the perturbation.

A.1. *Background*

The background is taken to be axisymmetric and aligned with the vertical axis. In cylindrical coordinates (R,ϕ,z), $\boldsymbol{v} = (v_R, v_\phi, v_z) = (0, R\Omega, 0)$. The angular frequency $\Omega(R,z)$ and $\lambda(R,z)$ are related by

$$R\Omega^2 = c_s^2\partial_R\lambda + \partial_R\Phi, \tag{A6}$$

$$0 = c_s^2\partial_z\lambda + \partial_z\Phi \tag{A7}$$

based on Equation (A1).

Taking ∂_z(A6) - ∂_R(A7) gives $\partial_z\Omega = 0$. Hence, Ω can be determined by evaluating Equation (A6) at the midplane, which yields

$$\Omega^2 = \Omega_{\rm K}^2 + \frac{c_s^2}{R}\partial_R\lambda_{\rm mid} \tag{A8}$$

where $\Omega_{\rm K} = \Omega_{\rm K}(R)$ is the Keplerian frequency and $\lambda_{\rm mid} = \lambda|_{z=0}$.

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Taking $z\times$(A6) - $R\times$(A7) leads to

$$\partial_z \lambda = -\frac{z}{H^2} + \frac{z}{R}\partial_R \lambda, \tag{A9}$$

where

$$H = c_s/\Omega\ . \tag{A10}$$

Near the midplane of the disk, the first term on the right-hand side is dominant over the second as $H^2 \ll R^2$, so $\rho \propto \exp\left\{-z^2/(2H^2)\right\}$. Therefore, the background volume density of the disk has the form

$$\rho(R,z) = \frac{\Sigma}{\sqrt{2\pi}H}\exp\left\{-\frac{z^2}{2H^2}\right\}, \tag{A11}$$

where $\Sigma(R)$ is the radial profile of the disk surface density, which can be chosen freely.

In terms of Σ, Equation (A8) becomes

$$\Omega^2 = \Omega_{\rm K}^2 + \frac{c_s^2}{R}\partial_R \ln\left(\frac{\Sigma}{H}\right). \tag{A12}$$

We shall also need the epicyclic frequency,

$$\kappa^2 \equiv R^{-3}\partial_R(R^4\Omega^2) \tag{A13}$$

and its deviation from Ω via

$$\epsilon \equiv \frac{\kappa^2}{\Omega^2} - 1 \tag{A14}$$

$$= R\partial_R \ln(\Omega^2 R^3) \tag{A15}$$

$$\approx R\partial_R\left(\frac{H^2}{R^2}R\partial_R \ln\frac{\Sigma}{H}\right) \tag{A16}$$

where the final expression is to leading order in H^2/R^2 when $d_R(\ln\Sigma)$ is order unity.

A.2. *Perturbation Equations*

Linearizing Equations (A1) and (A2) yields

$$\partial_t v_R' = -\Omega\partial_\phi v_R' + 2\Omega v_\phi' - c_s^2\partial_R\lambda', \tag{A17}$$

$$\partial_t v_\phi' = -\Omega\partial_\phi v_\phi' - (2 + R\partial_R \ln\Omega)\,\Omega v_R' - \frac{c_s^2}{R}\partial_\phi\lambda', \tag{A18}$$

$$\partial_t v_z' = -\Omega\partial_\phi v_z' - c_s^2\partial_z\lambda', \tag{A19}$$

$$\partial_t \lambda' = -\Omega\partial_\phi\lambda' - \frac{1}{R}\partial_R\left(Rv_R'\right) - (\partial_R\lambda)\,v_R' - \frac{1}{R}\partial_\phi v_\phi' - (\partial_z\lambda)\,v_z' - \partial_z v_z', \tag{A20}$$

Following Tanaka et al. (2002) and Ogilvie (2008), we take the azimuthal dependency of linearized variables to be $\propto e^{-i\phi}$, as is appropriate for a warp, and we decompose the vertical dependency in Hermite polynomials. We further simplify by restricting the Hermite expansion to the leading-order contributions, which results in setting

$$\frac{v_R'}{\Omega R} = U_R\frac{z}{H}e^{-i\phi}, \tag{A21}$$

$$\frac{v_\phi'}{\Omega R} = U_\phi\frac{z}{H}e^{-i\phi}, \tag{A22}$$

$$\frac{v_z'}{\Omega R} = U_z e^{-i\phi}, \tag{A23}$$

$$\lambda' = \Lambda\frac{z}{H}e^{-i\phi}, \tag{A24}$$

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where U_R, U_ϕ, U_z, and Λ are complex coefficients that depend on R and t.[6] Plugging these into Equations (A17) to (A20) gives the evolution equations

$$\Omega^{-1}\partial_t U_R = iU_R + 2U_\phi + h^2 R(\partial_R \ln H)\Lambda - h^2 R\partial_R \Lambda, \tag{A25}$$

$$\Omega^{-1}\partial_t U_\phi = iU_\phi - (2 + R\partial_R \ln \Omega)\, U_R + ih^2\Lambda, \tag{A26}$$

$$\Omega^{-1}\partial_t U_z = iU_z - h\Lambda, \tag{A27}$$

$$\Omega^{-1}\partial_t \Lambda = i\Lambda - R\partial_R \ln\left(\Omega\Sigma H R^2\right) U_R - R\partial_R U_R + iU_\phi + \frac{1}{h}U_z - hR\partial_R \ln\left(\Sigma H^2\right)U_z, \tag{A28}$$

where

$$h = H/R \tag{A29}$$

is the aspect ratio. To derive Equation (A28), we use Equations (A9) and (A11) to calculate $\partial_z\lambda$ and $\partial_R\lambda$, respectively. In addition, although the factors of z cancel from the first three equations, some of the terms in Equation (A28) have coefficient z, and others have coefficient z^3. Since we are really extracting the projection of this equation onto the first Hermite polynomial, we replace those coefficients by their projections, which amounts to replacing $z \to H$ and $z^3 \to 3H^3$.

A.3. *Steady-State Equations*

Henceforth, we consider steady-state equations ($\partial_t \to 0$). Equation (A27) gives $\Lambda = iU_z/h$, which can be used to eliminate Λ in other three equations. We then eliminate U_ϕ by forming two different combinations of the three equations: $-i\times$(A25)$+2\times$(A26) and $-i\times$(A25)$+$(A26)$+$(A28). These combinations gives

$$0 = \frac{d}{dR}\left[\Sigma H\Omega^2 R^3\left(U_R + hU_z\right)\right], \tag{A30}$$

$$0 = \frac{d\ln\left(\Omega^2 R^3\right)}{dR}\left(U_R + hU_z\right) + h\frac{d}{dR}U_z, \tag{A31}$$

which are two equations for two unknowns: $U_R + hU_z$ and U_z. The former is related to the radial speed in the (spherical) $\hat{\boldsymbol{r}}$ direction, $v_r' = (R/r)v_R' + (z/r)v_z'$, which implies

$$\frac{v_r'}{\Omega R} \approx \left(U_R + hU_z\right)\frac{z}{H}e^{-i\phi} \tag{A32}$$

after dropping the $O(z^2/r^2)$ correction. We therefore define

$$U_r \equiv U_R + hU_z \tag{A33}$$

which represents the amplitude of v_r'. Equations (A30-A31) then become

$$0 = \frac{d}{dR}\left(\frac{\sigma}{2h}U_r\right), \tag{A34}$$

$$0 = \epsilon U_r + hR\frac{d}{dR}U_z, \tag{A35}$$

where

$$\sigma \equiv 2\pi\Sigma H^2\Omega^2 R^2 \tag{A36}$$

as defined in Equation (8) in the main text and we used Equation (A15) to insert ϵ. We will get the final equations after adding an extra $-2i\alpha u_r$ term to the right-hand-side of Equation (A35) due to the effects of viscosity. We show in Appendix A.4 how to derive the viscous term.

[6] We let the physical field be the real part of their complex expression.

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A.4. *Viscous Terms*

Here, we derive the viscous term we added in the last step of the previous section. We take the viscous force per unit mass to be

$$\boldsymbol{f}_{\rm tot} = \frac{1}{R\Omega^2\rho_{\rm tot}}\boldsymbol{\nabla}\cdot(\nu\rho_{\rm tot}\boldsymbol{\tau}_{\rm tot}), \tag{A37}$$

where $\boldsymbol{\tau}$ is the stress tensor and ν is the kinematic viscosity,

$$\nu = \alpha c_s H, \tag{A38}$$

with α being the viscosity parameter as in the Shakura & Sunyaev (1973) model. This $\boldsymbol{f}_{\rm tot}$ is to be added to the right-hand side of Equation (A1).

We derive the perturbed $\boldsymbol{f}'$, which is to be inserted on the right-hand-side of the steady-state versions of Equations (A17)–(A19) by first working in the zero-inclination frame, where $v_z' = 0$. We shall then rotate to obtain the general form. Anticipating that the vertical viscous force is small, Equation (A19) implies $\lambda' = 0$. Then,

$$\boldsymbol{f}' = \frac{1}{R\Omega^2\rho}\boldsymbol{\nabla}\cdot(\nu\rho\boldsymbol{\tau}'). \tag{A39}$$

The most important terms in $\boldsymbol{\tau}'$ are

$$\tau_{Rz}' = \tau_{zR}' = \partial_z v_R', \tag{A40}$$
$$\tau_{\phi z}' = \tau_{z\phi}' = \partial_z v_\phi', \tag{A41}$$

as the main effect of viscosity is to act on the sloshing-induced vertical shear in the horizontal velocity components (see also Papaloizou & Lin 1995). Hence,

$$\boldsymbol{f}' \simeq \frac{1}{R\Omega^2\rho}\frac{\partial}{\partial z}\left[\nu\rho\left(\hat{\boldsymbol{R}}\partial_z v_R' + \hat{\boldsymbol{\phi}}\partial_z v_\phi'\right)\right] \tag{A42}$$
$$\simeq -\frac{\alpha z}{R\Omega}\left[\left(\partial_z v_R'\right)\hat{\boldsymbol{R}} + \left(\partial_z v_\phi'\right)\hat{\boldsymbol{\phi}}\right] \tag{A43}$$
$$\simeq -\frac{\alpha v_R'}{R\Omega}\left[\hat{\boldsymbol{R}} + \left(\frac{i}{2}\right)\hat{\boldsymbol{\phi}}\right], \tag{A44}$$

where we keep leading order terms by assuming $\partial_R, \partial_\phi \to \mathcal{O}(1)$ and $\partial_z \to \mathcal{O}(1/h)$ (see also Papaloizou & Lin 1995), and in the third equality we eliminate v_ϕ' by using the dominant contribution from Equation (A17).

In order to rotate $\boldsymbol{f}'$, we simply replace $v_R' \to v_r'$; other contributions to the rotation are higher order. The result is that Equations (A25)–(A26) are to be modified by adding to their respective right-hand-sides $-\alpha U_r$ and $\frac{1}{2}i\alpha U_r$. Finally, when we carry through the manipulations described in Section A.3, the viscous force adds $-2i\alpha U_r$ to the right-hand side of Equation (A35). It also adds a term to Equation (A34), but one that is smaller than the other in Equation (A34) term by $\mathcal{O}(\alpha)$, and so we drop it. Our viscous term is the same as those in Papaloizou & Lin (1995) and Lubow & Ogilvie (2000), except that their terms are proportional to U_R.

A.5. *Final Equations*

Adding the viscous term $-2i\alpha U_r$ to the right-hand side of Equation (A35), we obtain the steady-state equations

$$\frac{d}{dR}\left(\frac{\sigma}{2h}U_r\right) = 0, \tag{A45}$$
$$(2\alpha + i\epsilon)\,U_r = -ihR\frac{dU_z}{dR}. \tag{A46}$$

To better connect these equations to those in the main text, we define

$$U_\perp = \frac{U_r}{2h} \tag{A47}$$

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and get

$$\frac{d}{dR}\left(\sigma U_\perp\right) = 0, \tag{A48}$$

$$\left(4\alpha + 2i\epsilon\right) U_\perp = -iR\frac{dU_z}{dR}. \tag{A49}$$

In the next Appendix, we will show that these equations are equivalent to the vectorial equations, i.e., Equations (5) to (7) in the main text. The new quantity $U_\perp$ is related to the physical v_r field as

$$\frac{v_r'}{\Omega R} \approx 2U_\perp \frac{z}{R} e^{-i\phi}, \tag{A50}$$

obtained by replacing $U_r \equiv U_R + hU_z$ in Equation (A32) with $U_\perp$. This equation is provided in the main text as Equation (12)

B. VECTORIAL LINEAR EQUATIONS

Now we derive the vectorial steady state equations in Section 3.3, i.e., Equations (5) to (7). This can be done by converting Equations (A48) and (A49) into their corresponding vectorial form. The derivation takes two steps. First, we show that there is one-to-one relation to replace the complex numbers $\{U_z, U_\perp, \sigma U_\perp\}$ by vectors $\{\hat{\boldsymbol{l}}, \boldsymbol{U}_\perp, \boldsymbol{G}_\perp\}$. Then, we replace the complex numbers in Equations (A48) and (A49) by their vectorial counterparts to get Equations (5) to (7).

B.1. *One-to-one Relations*

We first show that

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \hat{\boldsymbol{l}} = -iU_z, \tag{B51}$$

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{G}_\perp = -\sigma U_\perp, \tag{B52}$$

in the linear regime. We calculate the left-hand sides of the two equations using the integral definition of $\boldsymbol{L}$ and $\boldsymbol{G}_\perp$, given by Equations (1) and (14). Let the density and velocity in the integrals be the sum of their background values and small perturbations, i.e., $\rho_{\rm tot} = \rho(1 + \lambda')$ and $\boldsymbol{v}_{\rm tot} = \boldsymbol{v} + \boldsymbol{v}'$, we find that the integrals become

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{L} = \Omega R^3 \int_0^{2\pi}\int_{-\infty}^{+\infty} \left(-i\frac{v_z'}{\Omega R} + i\frac{z}{R}\frac{v_R'}{\Omega R} - \frac{z}{R}\frac{v_\phi'}{\Omega R} - \lambda'\frac{z}{R}\right) e^{i\phi}\rho dz d\phi, \tag{B53}$$

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{G}_\perp = -\Omega R^3 \int_0^{2\pi}\int_{-\infty}^{+\infty} \frac{z}{R} v_r' e^{i\phi}\rho dz d\phi. \tag{B54}$$

Here we dropped the nonlinear perturbative terms, and we switch from spherical to cylindrical integral, introducing a negligible $\mathcal{O}(\beta^2)$ error with the $\beta \ll 1$ being the disk inclination. Note that the right-hand sides of Equations (B53) and (B54) are Fourier-Hermite integrals, so we have

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{L} = \pi\Sigma\Omega R^3 \left[-iU_z + h\left(iU_R - U_\phi - \Lambda\right)\right], \tag{B55}$$

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{G}_\perp = -\pi\Sigma H R^3 \Omega^2 U_r, \tag{B56}$$

where U_z, U_R, U_ϕ, Λ and U_r are the Fourier-Hermite coefficients of the perturbations as defined in Appendix A.2.

For Equation (B55), we note that (i) $h\Lambda = iU_z$ in steady state, and (ii) the hU_R and hU_ϕ terms are small comparing to U_z as long as $h \lesssim |2\alpha + i\epsilon|$ (i.e., the non-resonant condition, see Ogilvie 1999). Therefore, Equation (B55) gives

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{L} = 2\pi\Sigma\Omega R^3(-iU_z). \tag{B57}$$

Equation (B51) follows directly because $\boldsymbol{L} = 2\pi\Sigma\Omega R^3\hat{\boldsymbol{l}}$.

For Equation (B56), we plug in $U_\perp \equiv U_r/(2h)$ and $\sigma \equiv 2\pi\Sigma H^2\Omega^2R^2$ to get Equation (B52).

We now define the vector $\boldsymbol{U}_\perp \equiv -\boldsymbol{G}_\perp/\sigma$ to get the final relation,

$$\left(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}\right) \cdot \boldsymbol{U}_\perp = U_\perp, \tag{B58}$$

from Equation (B56).

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B.2. *Main Steady-state Equations*

In Equations (A48) and (A49), we replace U_z, $U_\perp$, and $\sigma U_\perp$ using Equations (B51), (B52), and (B58), and we replace all R by r. These substitutions are valid with a factor of $\mathcal{O}(\beta^2)$ error. The complex Equations (A48) and (A49) becomes

$$(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot \frac{d}{dr}\boldsymbol{G}_\perp = 0 \tag{B59}$$

$$4\alpha(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot \boldsymbol{U}_\perp + 2\epsilon\,(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot \left(\hat{\boldsymbol{l}} \times \boldsymbol{U}_\perp\right) = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot \left(r\frac{d}{dr}\hat{\boldsymbol{l}}\right) = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot \boldsymbol{\psi}, \tag{B60}$$

where in the second line we have used the relation $i(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \times \hat{\boldsymbol{l}}$, the scalar triple product identity $[(\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \times \hat{\boldsymbol{l}}] \cdot \boldsymbol{U}_\perp = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot (\hat{\boldsymbol{l}} \times \boldsymbol{U}_\perp)$, and the definition of $\boldsymbol{\psi} \equiv r d\hat{\boldsymbol{l}}/dr$.

Taking $\hat{\boldsymbol{x}} \times$ (real part) $+ \hat{\boldsymbol{y}} \times$ (imaginary part) for Equations (B59) and (B60), we get

$$\frac{d}{dr}\boldsymbol{G}_\perp = 0, \tag{B61}$$

$$4\alpha\boldsymbol{U}_\perp + 2\epsilon\hat{\boldsymbol{l}} \times \boldsymbol{U}_\perp = \boldsymbol{\psi}. \tag{B62}$$

Together with the definition $\boldsymbol{U}_\perp \equiv -\boldsymbol{G}_\perp/\sigma$ motivated in the last subsection, we have derived the main steady-state conditions Equations (5) to (7).

B.3. *Other Equations*

Here, we briefly discuss three more relations that are used in linear Sections of the main text.

From Equations (3) and (B51), we have

$$-iU_z = \beta e^{i\gamma} \tag{B63}$$

to linear order in the disk inclination. Thus, $-iU_z$ is equal to the complex inclination. Physically, when a disk is tilted away from the $z = 0$ plane, part of the orbital motion is projected onto the $\hat{\boldsymbol{z}}$-direction. That is why the vertical velocity perturbation U_z is directly related to the inclination. The factor i comes from phase offset: the vertical velocity v_z is maximal when a fluid element crosses the reference plane, and becomes zero when the element reaches its maximum vertical displacement.

Following the meaning of $-iU_z$, and from Equation (B51), it can be seen that the complex warp is given by

$$\psi \equiv (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}) \cdot \boldsymbol{\psi} = -iR\frac{dU_z}{dR}. \tag{B64}$$

Plugging this into Equation (A49), we get

$$(4\alpha + 2i\epsilon)U_\perp = \psi \quad \text{or} \quad U_\perp = \frac{\psi}{4\alpha + 2i\epsilon}, \tag{B65}$$

which is used in the main text as Equation (13)

The vectorial version of Equation (13) can be obtained by taking $[2\alpha$(B62)-$\epsilon\hat{\boldsymbol{l}}\times$(B62)$]/(4\alpha^2 + \epsilon^2)$, which gives

$$\boldsymbol{U}_\perp = \frac{\alpha}{4\alpha^2 + \epsilon^2}\boldsymbol{\psi} \;-\; \frac{\epsilon}{2\,(4\alpha^2 + \epsilon^2)}\hat{\boldsymbol{l}} \times \boldsymbol{\psi}. \tag{B66}$$

This relation is given as Equation (15) in the main text.

C. NONLINEAR THEORY

The purpose of this Appendix is to present a self-contained derivation of the nonlinear relations used in our main text. We begin with the exact shell-integrated conservation laws, then derive the expression for the accretion rate $\dot{M}$, motivate our approximate steady-state condition for the in-plane torque $\boldsymbol{G}_\perp$, and finally present a local stability argument for the breaking instability.

We follow the framework of Ogilvie & Latter (2013a) and consider the conservation of mass and angular momentum for a spherical shell:

$$\partial_t\mathcal{M} + \partial_r\dot{M} = 0, \tag{C67}$$

$$\partial_t\boldsymbol{L} + \partial_r\boldsymbol{G}_{\rm tot} = 0. \tag{C68}$$

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Here $\mathcal{M}$ and $\boldsymbol{L}$ are the mass and angular momentum per unit radius, defined as

$$\mathcal{M} = \int \rho r^2 d\Omega = 2\pi r \Sigma, \tag{C69}$$

$$\boldsymbol{L} = \int \rho \left(\boldsymbol{r} \times \boldsymbol{v}\right) r^2 d\Omega, \tag{C70}$$

with $d\Omega \equiv \sin\theta d\theta d\phi$. The terms $\dot{M}$ and $\boldsymbol{G}_{\rm tot}$ are the radial flux of mass and angular momentum,

$$\dot{M} = \int v_r \rho r^2 d\Omega, \tag{C71}$$

$$\boldsymbol{G}_{\rm tot} = \int v_r \rho \left(\boldsymbol{r} \times \boldsymbol{v}\right) r^2 d\Omega + \boldsymbol{F}_\nu, \tag{C72}$$

with $\boldsymbol{F}_\nu$ denoting the viscous contribution.

It is convenient to rewrite the conservation laws in terms of the shell-averaged specific angular momentum,

$$\boldsymbol{h} = \frac{\boldsymbol{L}}{\mathcal{M}} \tag{C73}$$

Subtracting $\boldsymbol{h}$ times Equation (C67) from Equation (C68) gives

$$\mathcal{M}\partial_t \boldsymbol{h} + \dot{M}\partial_r \boldsymbol{h} + \partial_r \boldsymbol{G} = 0, \tag{C74}$$

where

$$\boldsymbol{G} = \boldsymbol{G}_{\rm tot} - \dot{M}\boldsymbol{h} \tag{C75}$$

is the internal torque, obtained by subtracting the angular momentum flux driven by the mass accretion flow $\dot{M}$ from the total flux $\boldsymbol{G}_{\rm tot}$.

It will soon turn out to be useful to express $\boldsymbol{G}$ in terms of its spacial components. Following Ogilvie & Latter (2013a), we parameterize $\boldsymbol{G}$ using dimensionless coefficients Q's as

$$\boldsymbol{G} = -2\sigma\left(Q_1\hat{\boldsymbol{l}} + Q_2\boldsymbol{\psi} + Q_3\hat{\boldsymbol{l}} \times \boldsymbol{\psi}\right), \tag{C76}$$

where

$$\hat{\boldsymbol{l}} \equiv \frac{\boldsymbol{L}}{|\boldsymbol{L}|}, \quad \boldsymbol{\psi} \equiv \frac{\partial \hat{\boldsymbol{l}}}{\partial \ln r}, \quad \text{and} \quad \sigma \equiv \pi \Sigma r^2 H^2 \Omega^2, \tag{C77}$$

as defined in the main text. In warp steady states, the values of Q's are functions of $|\boldsymbol{\psi}|$, α, and ϵ, and can be calculated using the shear-box theory in Ogilvie & Latter (2013a).

C.1. *Accretion Flow* $\dot{M}$

Now we can derive Equations (24) and (25), namely the expression for the accretion rate in a warped disk. Projecting Equation (C74) onto the $\hat{\boldsymbol{l}}$ direction gives

$$\dot{M}\partial_r h + \hat{\boldsymbol{l}} \cdot \partial_r \boldsymbol{G} = 0, \tag{C78}$$

where $h = |\boldsymbol{h}|$. We have let $\partial_t h = 0$ to get the equation above because, for a nearly-Keplerian disk, $h = r^2\Omega$ is determined by the background rotation curve and can be considered a constant of time. Using $\hat{\boldsymbol{l}}\cdot\partial_r\boldsymbol{G} = \partial_r(\hat{\boldsymbol{l}}\cdot\boldsymbol{G}) - \boldsymbol{G}\cdot\partial_r\hat{\boldsymbol{l}}$ and $\partial_r\hat{\boldsymbol{l}} = \boldsymbol{\psi}/r$, we find

$$\dot{M}\partial_r h = \partial_r\left(\hat{\boldsymbol{l}} \cdot \boldsymbol{G}\right) - \frac{1}{r}\sigma \boldsymbol{G} \cdot \boldsymbol{\psi}, \tag{C79}$$

which is given as Equation (24) in the main text. Substituting Equation (C76) then gives

$$\dot{M}\partial_r h = 2\partial_r\left(\sigma Q_1\right) - \frac{2}{r}\sigma Q_2 |\boldsymbol{\psi}|^2, \tag{C80}$$

which is the Equation (25) in the main text. To the linear order of $\boldsymbol{\psi}$, both Equations (C79) and (C80) will reduce to the standard $\dot{M}$ expression for flat α disk. When $|\boldsymbol{\psi}| > \alpha$, $\dot{M}$ is mainly driven by $Q_2|\boldsymbol{\psi}|^2$ and the standard α driven accretion rate becomes negligible, as we discussed in Section 6.4.

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C.2. *Constancy of the In-plane Torque* $\boldsymbol{G}_\perp$

We now motivate the approximate steady-state condition used Section 5 for the in-plane torque,

$$\partial_r |\boldsymbol{G}_\perp|^2 = 0, \tag{C81}$$

where

$$\boldsymbol{G}_\perp = -2\sigma(Q_2\boldsymbol{\psi} + Q_3\hat{\boldsymbol{l}} \times \boldsymbol{\psi}) \tag{C82}$$

as we can read from Equation (C76).

In the linear theory, $\partial_r \boldsymbol{G}_\perp = 0$ is the exact steady-state condition. In the nonlinear regime, however, $\boldsymbol{G}_\perp$ changes direction with radius because $\hat{\boldsymbol{l}}$ itself changes direction, so $\partial_r \boldsymbol{G}_\perp = 0$ no longer holds. The natural nonlinear generalization is to track the variation of the magnitude $|\boldsymbol{G}_\perp|$.

Since $|\boldsymbol{G}_\perp|^2 = \boldsymbol{G}_\perp \cdot \boldsymbol{G}_\perp$, its radial derivative is

$$\partial_r \left|\boldsymbol{G}_\perp\right|^2 = 2\boldsymbol{G}_\perp \cdot \left\{\partial_r \boldsymbol{G} - \partial_r \left[(\boldsymbol{G}\cdot\hat{\boldsymbol{l}})\hat{\boldsymbol{l}}\right]\right\}. \tag{C83}$$

Using Equation (C74) to eliminate $\boldsymbol{G}$ and plugging $\boldsymbol{G}\cdot\hat{\boldsymbol{l}} = -\sigma Q_1$ into the equation above, we have

$$\partial_r \left|\boldsymbol{G}_\perp\right|^2 = 2\boldsymbol{G}_\perp \cdot \left(-\mathcal{M}h\partial_t\hat{\boldsymbol{l}} - \frac{\dot{M}h}{r}\boldsymbol{\psi} + \frac{2\sigma Q_1}{r}\boldsymbol{\psi}\right), \tag{C84}$$

where the terms in the parentheses represent the sources of torque variation. This is equation is exact, except for the only approximation being $\partial_t h = 0$. It is the more complete form of the correction quoted as Equation (20) in Section 6.2.

In the moderately warped regime where $|\boldsymbol{\psi}| > \alpha$, the Q_1 term is typically smaller than the $\dot{M}$ term, so that we have

$$\partial_r \left|\boldsymbol{G}_\perp\right|^2 \simeq 2\boldsymbol{G}_\perp \cdot \left(-\mathcal{M}h\partial_t\hat{\boldsymbol{l}} - \frac{\dot{M}h}{r}\boldsymbol{\psi}\right) \tag{C85}$$

as Equation (20). Equation (C80) further shows that $\dot{M}h\boldsymbol{\psi}/r$ is of order $|\boldsymbol{G}_\perp||\boldsymbol{\psi}|^2$. Therefore, Equation (C85) implies

$$\frac{\partial}{\partial \ln r} \ln \left|\boldsymbol{G}_\perp\right|^2 \simeq O(|\boldsymbol{\psi}|^2), \tag{C86}$$

so long as the residual time dependence is also small. This motivates the approximation

$$\partial_r |\boldsymbol{G}_\perp|^2 \simeq 0 \tag{C87}$$

as the steady-state condition adopted in Section 5. It is not exact, but it remains a useful approximation over the range of warp amplitudes explored by in this work.

C.3. *Stability of Steady-state Solutions*

We have shown in Section 5.2 how to get solutions to the nonlinear warp equations. Now we analyze the stability of these solutions. Let $\boldsymbol{\psi}_0$ be a steady-state warp and consider a small perturbation $\delta\boldsymbol{\psi}$ parallel to $\boldsymbol{\psi}_0$, i.e.,

$$\boldsymbol{\psi} = \boldsymbol{\psi}_0 + \delta\boldsymbol{\psi}. \tag{C88}$$

We focus on nonlinear warps so $\boldsymbol{\psi}_0$ can be large, but we will perform a linear stability analysis so we only consider linear-order terms of $\delta\psi \equiv |\delta\boldsymbol{\psi}|$. That gives the perturbation of the internal torque as

$$\delta\boldsymbol{G} = -2\delta\psi\sigma\left[\partial_{|\boldsymbol{\psi}|}Q_1\hat{\boldsymbol{l}} + \partial_{|\boldsymbol{\psi}|}\left(Q_2\boldsymbol{\psi}\right) + \partial_{|\boldsymbol{\psi}|}\left(Q_3\hat{\boldsymbol{l}}\times\boldsymbol{\psi}\right)\right] \tag{C89}$$

to the linear order.

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For simplicity, we take $\delta\psi \propto e^{\gamma t + ikr}$, with k large enough for a local WKB approximation to be valid. Perturbing ∂_r (Equation C74) gives

$$\frac{\mathcal{M}h}{r}\partial_t\delta\psi + \frac{\dot{M}h}{r}\partial_r\delta\psi - 2\sigma\partial_{|\boldsymbol{\psi}|}\left(Q_2|\boldsymbol{\psi}|\right)\partial_r^2\left(\delta\psi\right) = 0, \tag{C90}$$

where we have inserted Equation (C89) for $\delta\boldsymbol{G}$, dropped ∂_r of the background terms, and keep only the coefficients in the direction of $\delta\boldsymbol{\psi}$. The perturbative equation simplifies to

$$\frac{\mathcal{M}h}{r}\gamma + \frac{ik\dot{M}}{r} + 2k^2\sigma\partial_{|\boldsymbol{\psi}|}\left(Q_2|\boldsymbol{\psi}|\right) = 0. \tag{C91}$$

The middle term is purely imaginary so it only affects the phase speed of the perturbation. The growth rate of the perturbation is controlled by the real terms,

$$\mathrm{Re}(\gamma) \propto -\partial_{|\boldsymbol{\psi}|}\left(Q_2|\boldsymbol{\psi}|\right). \tag{C92}$$

Therefore, the perturbation vanishes when $\partial_{|\boldsymbol{\psi}|}\left(Q_2|\boldsymbol{\psi}|\right) > 0$ and grows when $\partial_{|\boldsymbol{\psi}|}\left(Q_2|\boldsymbol{\psi}|\right) < 0$.

In Section 5, we have also assumed the disk to be nearly-Keplerian; in that regime, $Q_2 \gg Q_3$ so $Q_2|\boldsymbol{\psi}| = |\boldsymbol{U}_\perp|$. Therefore, we state that the steady-state solutions on the increasing branch of the $|\boldsymbol{U}_\perp|$-$|\boldsymbol{\psi}|$ are stable, but those entering the declining branch are susceptible to a runaway instability.

D. DETAILS OF SIMULATION

D.1. *Grid and Boundary Conditions*

Our simulations are preformed with Athena++ (Stone et al. 2020). We adopt spherical polar coordinates and uniformly spaced grid cells, with the azimuthal angle ϕ ranges from 0 to 2π and the radial r and polar θ coverages shown in Table 1. The ϕ boundary is periodic; two layer of ghost cells are attached to each r and θ boundary to implement boundary conditions.

The θ cutoffs are placed sufficiently far from the disk midplane ($\geq 10H$ away from at $r = R_0$, $\geq 5.5H$ at $r = 1.5R_0$, and $\geq 2.5H$ at $r = 3.5R_0$) to ensure they do not influence the disk evolution. We impose reflective boundary conditions, which are found to be the best for maintaining the vertical hydrostatic equilibrium of the disk and minimizing spurious inflows or outflows. Specifically, we copy the values of density ρ, radial velocity v_r, and azimuthal velocity v_ϕ from the last active cells into the ghosts, while the polar velocity v_θ is copied with its sign reversed. Similar boundary conditions have been used in a number of previous studies (e.g., Zhu 2019; Kimmig & Dullemond 2024).

The radial boundary condition needs to achieve two goals: (i) keeping the disk at fixed tilts, and (ii) allowing the gas flow freely. We denote quantities in the first/last active cells with $'$ and those in the ghost cells with $''$. For goal (i), we hold the density in the ghost cells at their initial values, i.e.,

$$\rho'' = \rho''(t = 0). \tag{D93}$$

For (ii), we set the velocity in ghost cells in three steps: (1) calculate the cylindrical velocity components (v'_R, v'_ϕ, v'_z) with $\hat{\boldsymbol{z}}$ pointing at the direction of the disk plane (defined by $\beta = 0$ and $\beta = \beta_{\rm out}$ at the inner and outer boundaries, respectively); (2) calculate these components for the ghost cells as

$$v''_R = v'_R\left(\frac{R''}{R'}\right)^{-1/2}, \tag{D94}$$

$$v''_\phi = v'_\phi\left(\frac{R''}{R'}\right)^{-1/2}, \tag{D95}$$

$$v''_z = 0; \tag{D96}$$

then (3) convert them to the spherical-polar components $(v''_r, v''_\theta, v''_\phi)$ and assign these values to the ghost cells.

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D.2. *Initial Condition and Tilting*

The initial disk is set in two steps: first set up a disk in its flat background state, and then form a warped structure by tilting the disk differentially. We use an initial background surface density profile

$$\Sigma(r) = \Sigma_0 r^{-3/2} \frac{1}{f_{\rm gap}(r)}, \tag{D97}$$

where

$$f_{\rm gap}(r) = 1 + \frac{K-1}{2}\left[\tanh\left(\frac{r-R_a}{\Delta_a}\right) - \tanh\left(\frac{r-R_b}{\Delta_b}\right)\right] \tag{D98}$$

controls the shape of the gap. The parameters R_a and R_b are the locations of the inner and outer gap edges, Δ_a and Δ_b are the steepness of the edges, and the constant K determines the depth of the gap. We use parameter values $(R_a,\ R_b,\ \Delta_a,\ \Delta_b,\ K) = (1,\ 2,\ 0.1,\ 0.2,\ 19)$ in all simulations. The background volume density ρ and Ω are set as in Appendix A.1.

To apply a radially varying tilting profile

$$\hat{\boldsymbol{l}} = [\sin(\beta)\cos(\gamma), \sin(\beta)\sin(\gamma), \cos(\beta)], \tag{D99}$$

we rotate the fluid element at $\boldsymbol{r} = (r, \theta, \phi)$ to a new coordinate $\boldsymbol{r}' = (r, \theta', \phi')$ so that

$$\boldsymbol{r}' = \mathbf{R}_z(\gamma)\mathbf{R}_y(\beta)\boldsymbol{r}, \tag{D100}$$

where $\mathbf{R}_z(\gamma)$ and $\mathbf{R}_y(\beta)$ are the standard 3D rotational matrices around the $\hat{\boldsymbol{z}}$ and $\hat{\boldsymbol{y}}$ axes, respectively. When we calculate $\boldsymbol{r}'$, we convert $\boldsymbol{r}$ into cartesian coordinate so that apply $\mathbf{R}_z(\gamma)$ and $\mathbf{R}_y(\beta)$ can be applied; the resulting $\boldsymbol{r}'$ is also in Cartesian coordinates, based on which we get spherical-polar coordinates (r, θ', ϕ'). As we rotate a fluid element from $\boldsymbol{r}$ to $\boldsymbol{r}'$, we change its velocity vector to

$$\boldsymbol{v}' = \mathbf{R}_z(\gamma)\mathbf{R}_y(\beta)\boldsymbol{v} \tag{D101}$$

and preserve its density ρ.

When we analyze a simulation, we can reverse this tilting procedure using the real-time measurement of β and γ. This allows us to map from the code coordinates $\boldsymbol{r}_{\rm sim}$ to the disk-plane coordiantes $\boldsymbol{r}_{\rm disk}$ via

$$\boldsymbol{r}_{\rm disk} = \mathbf{R}_y(-\beta)\mathbf{R}_z(-\gamma)\boldsymbol{r}_{\rm sim}. \tag{D102}$$

The disk-plane snapshots in Figures 5 and 12 are obtained this in way.

D.3. *Viscosity*

The public version of Athena++ applies constant kinematic viscosity ν by default. We implement the α viscosity prescription by manually enrolling a spatially varying ν in the problem generator.[7]

D.4. *Integration Scheme*

Local-Lax-Friedrichs method (LLF) is adopted to solve the hydrodynamic equations in our simulations, where the spatial reconstruction of the primitive fluid quantities is done using the piecewise Linear Method (PLM). We perform time integrations using the second-order accurate Runge-Kutta/Heun's method (RK2). All of the numerical methods here are already implemented in the public version of Athena++.[8]

At runtime, the simulation domains are typically divided into $4\times8\times13$ meshblocks in r, θ and ϕ directions, processed in parallel on 416 cores using MPI. Each simulation typically takes a 24 hours walltime to integrate for 150 code-unit time.

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[7] See details on Athena++'s discussion forum: https://github.com/PrincetonUniversity/athena/discussions/411.

[8] We thank Dr. Alexander Dittmann for his suggestions on the integration scheme, which boost the numerical stability of our simulations.

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